

Basic Cell Biology in Nervous System

實驗神經概論

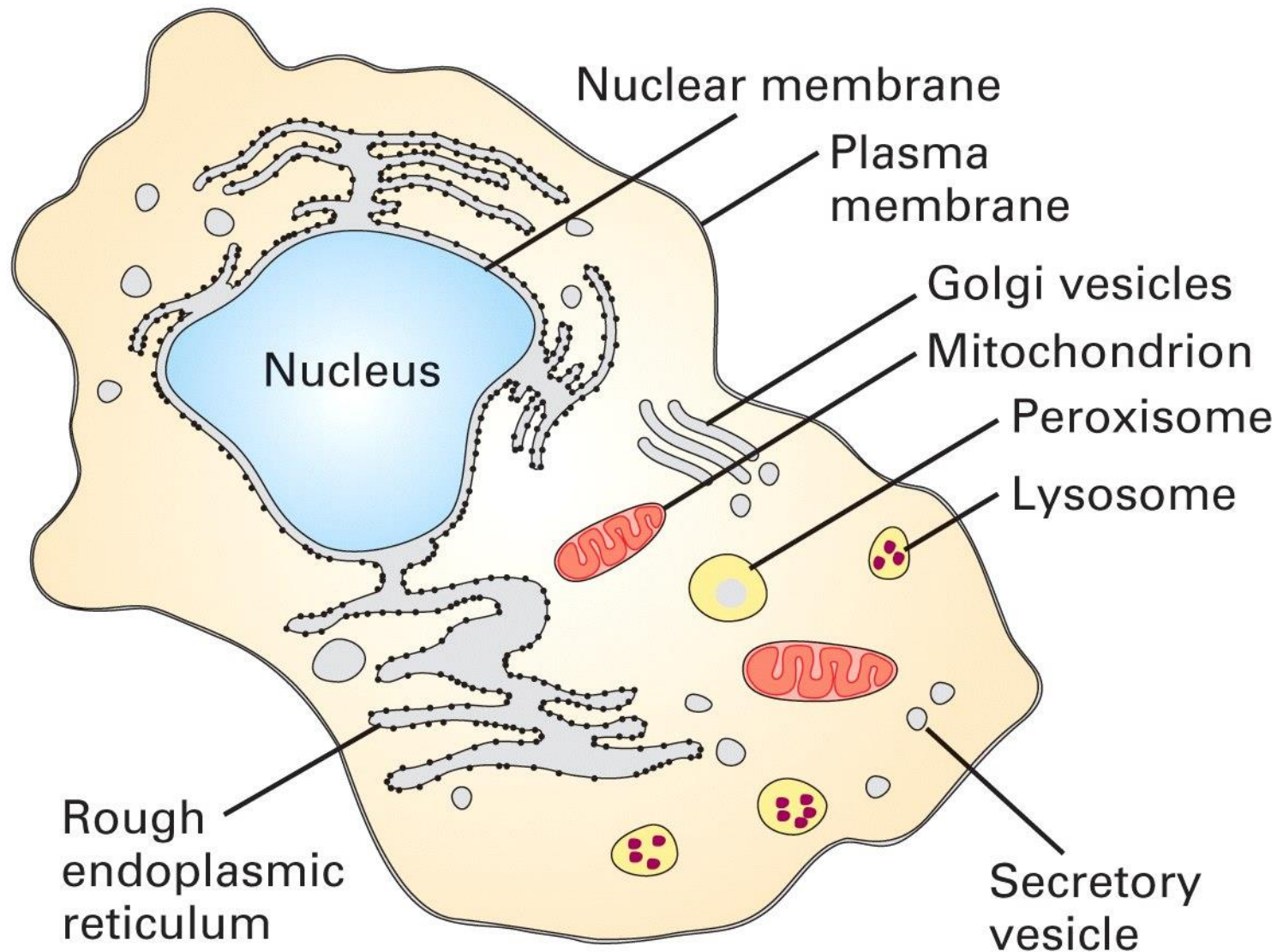
楊定一

2026/9/15

A simplified diagram of a eukaryotic cell.

FOUNDATION

(b) Eukaryotic cell



Principles of Neural Science

6th ed. (2021) — Kandel, Koester, Mack & Siegelbaum

**Primary source: Chapter 7 — The Cells of the
Nervous System**

*This lecture emphasizes core concepts. Slides marked
REFERENCE are supplementary and not core learning targets.*

Neuronal diversity: morphology and basic structure

Cell types in nervous systems

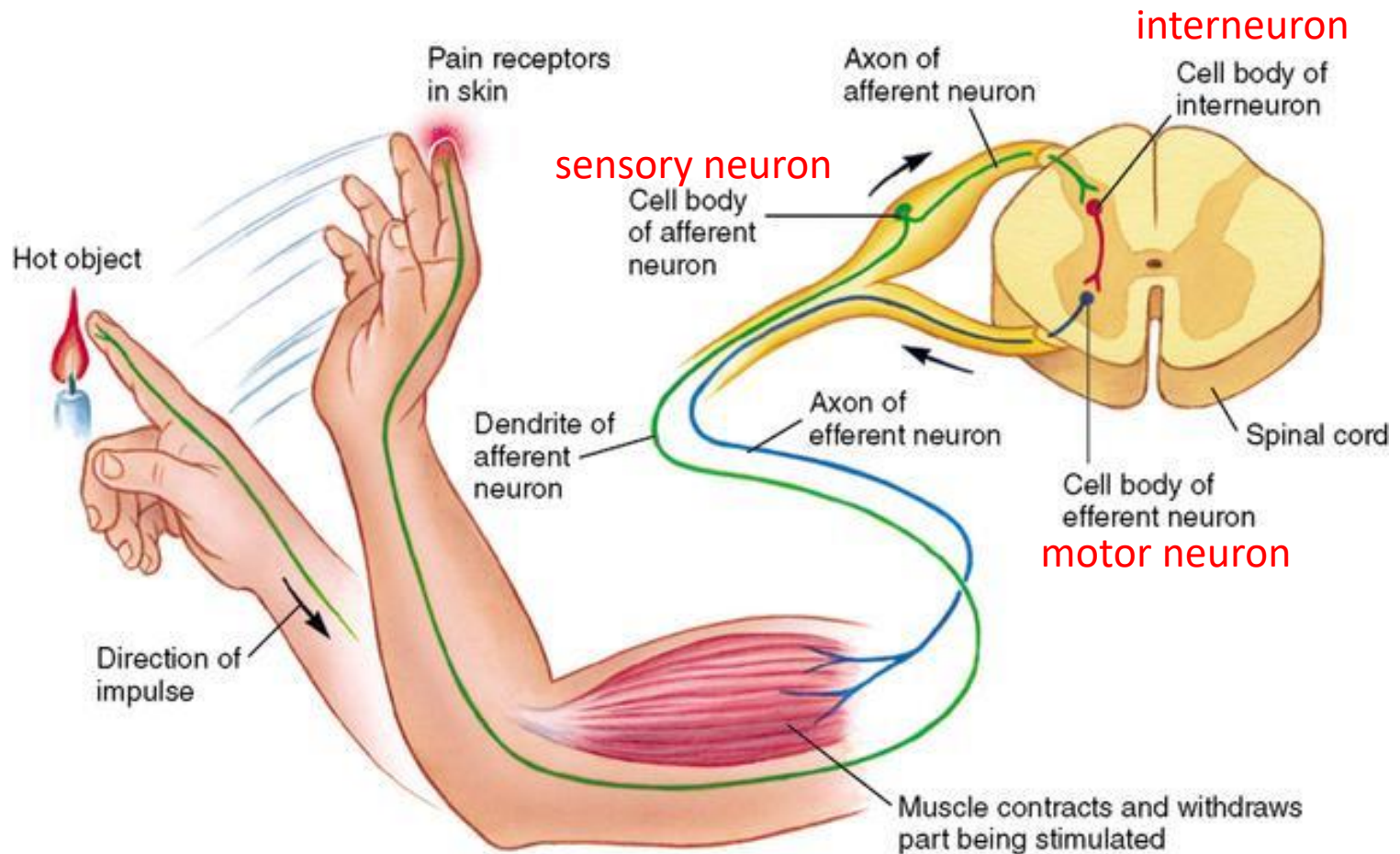
- **Neuron**

- **Sensory (afferent):** carry information from sensory receptors toward the CNS
- **Motor (efferent):** carry commands from the CNS to effectors such as muscle
- **Interneurons:** connect neurons and integrate signals within the CNS

- **Glia**

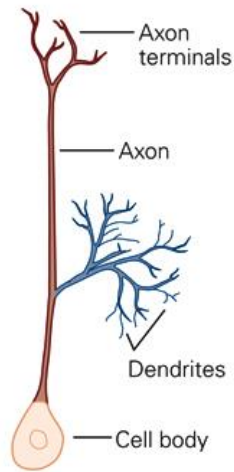
- **CNS:** astrocytes, oligodendrocytes, ependymal cells, microglia
- **PNS:** Schwann cells and satellite cells

Functional classification of three neuronal types



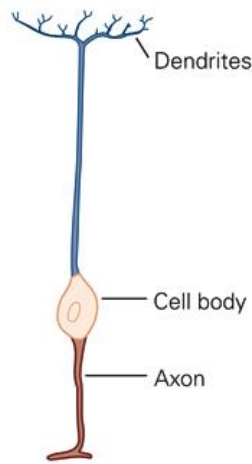
afferent = toward CNS
efferent = away from CNS

A Unipolar cell



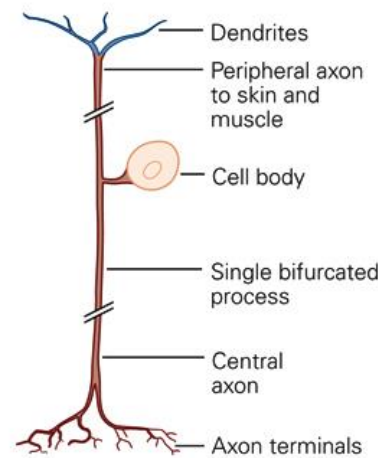
Invertebrate neuron

B Bipolar cell



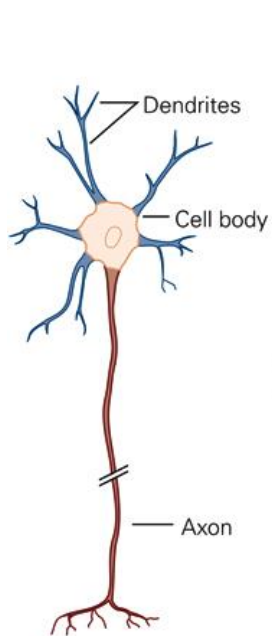
Bipolar cell of retina

C Pseudo-unipolar cell

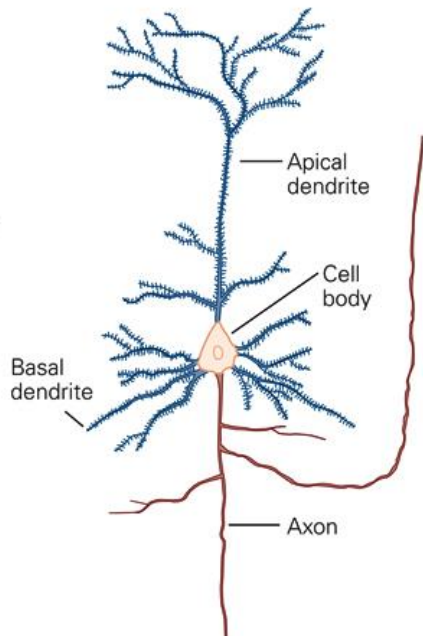


Ganglion cell of dorsal root

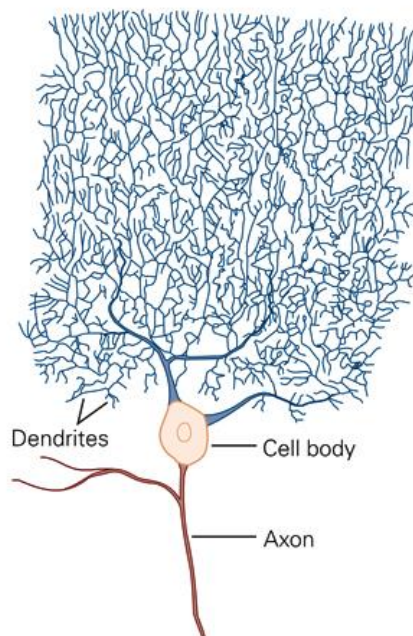
D Three types of multipolar cells



Motor neuron of spinal cord



Pyramidal cell of hippocampus



Purkinje cell of cerebellum

Morphology

A. Unipolar: one process

B. Bipolar: one axon + one dendrite

C. Pseudounipolar: one process that bifurcates into peripheral and central branches

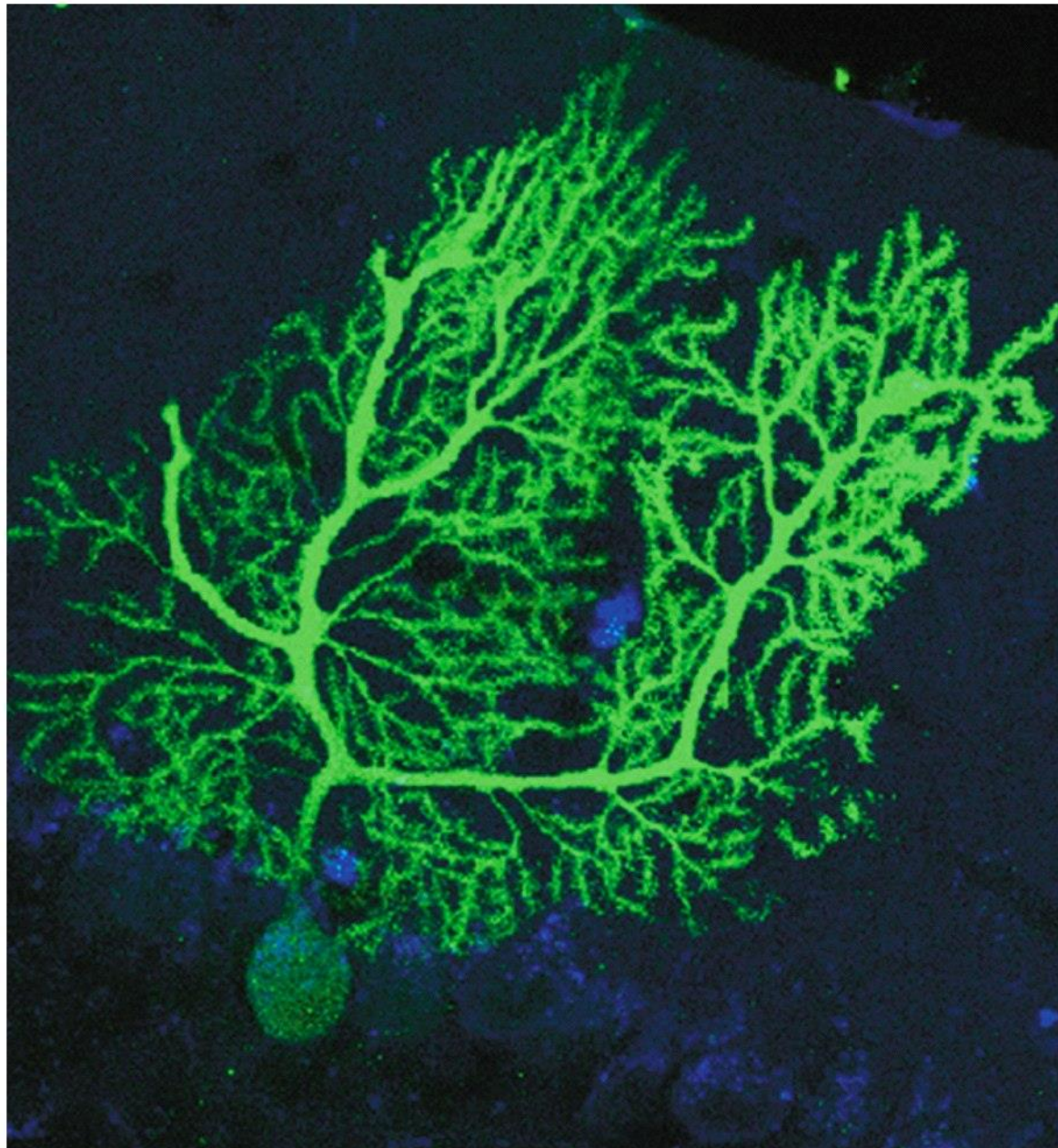
D. Multipolar: one axon + multiple dendrites

Functional direction

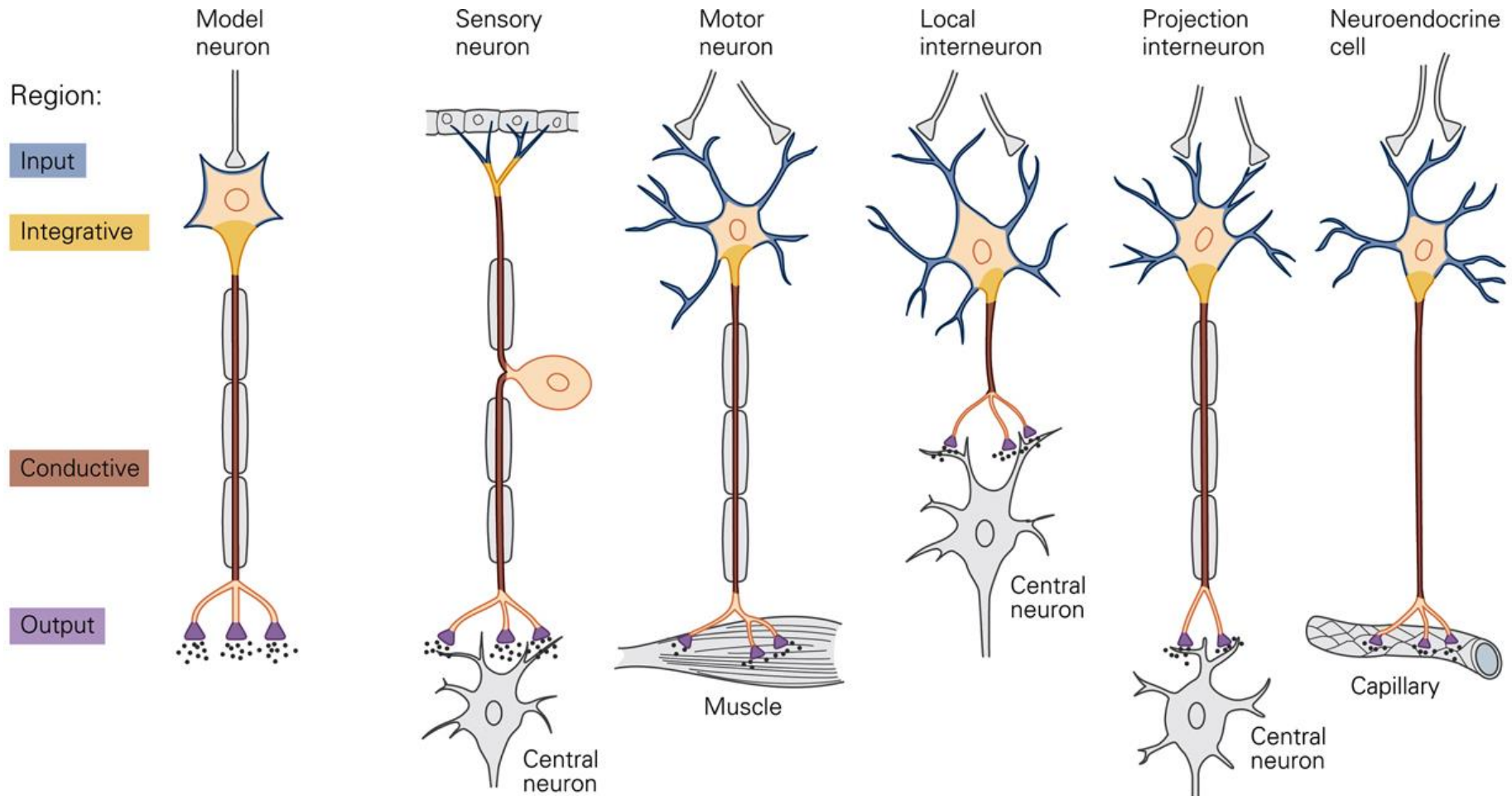
Afferent = toward the CNS (sensory)

Efferent = away from the CNS to effectors (motor)

Interneurons = local connections within the CNS



Purkinje neuron: an example of extreme dendritic complexity. 8

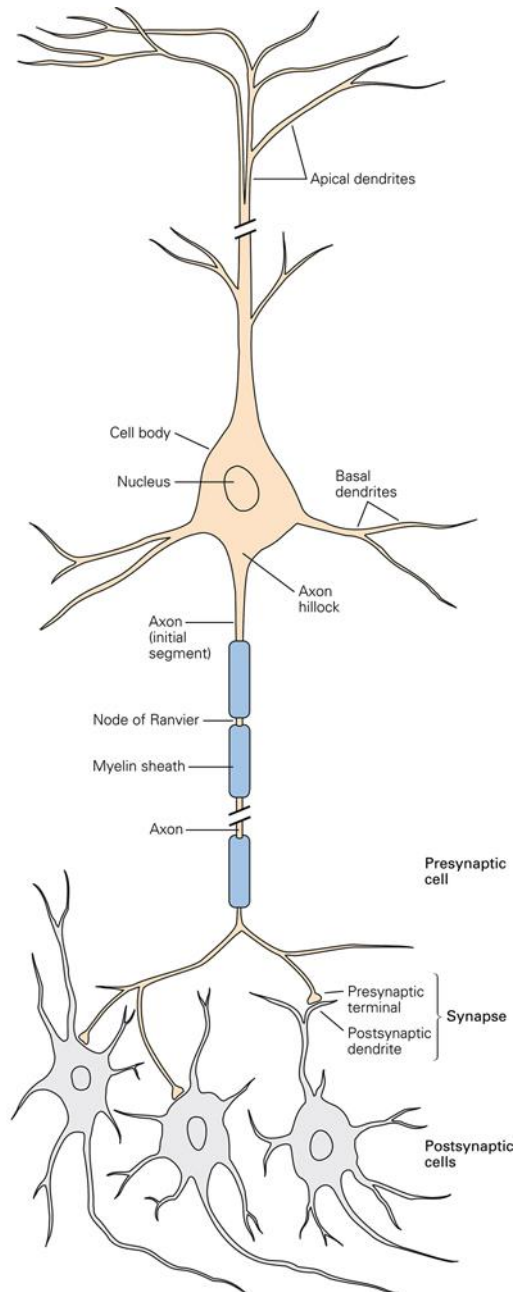


Most neurons can be described by four functional regions: **input → integration → conduction → output**.

Input, integration, and conduction are primarily electrical; output at chemical synapses is transmitter release. Not every neuron contains all four regions (for example, local interneurons).

Four Basic Structural Components of Neurons

- **Cell body (soma):** contains the nucleus and is the major site of protein and membrane synthesis.
- **Dendrites:** major input regions; receive synaptic signals and can support local protein synthesis.
- **Axon:** specialized for conduction of **action potentials** over long distances.
- **Axon terminals:** form synapses and release neurotransmitters or neuropeptides.



- Dendrites and the soma are the principal **input regions** of most neurons.
- The **axon initial segment** is the usual initiation site of action potential.
- The axon propagates electrical signals to **presynaptic terminals**, where chemical signaling occurs.
- In many axons, **myelin and nodes of Ranvier** support rapid conduction.

Chapter 07. The Cells of the Nervous System

- 7-1. Neurons and Glia Share Many Structural and Molecular Characteristics
- 7-2. The Cytoskeleton Determines Cell Shape
- 7-3. Protein Particles and Organelles Are Actively Transported Along the Axon and Dendrites
 - 7-3-1. Fast Axonal Transport Carries Membranous Organelles
 - 7-3-2. Slow Axonal Transport Carries Cytosolic Proteins and Elements of the Cytoskeleton
- 7-4. Proteins Are Made in Neurons as in Other Secretory Cells
 - 7-4-1. Secretory and Membrane Proteins Are Synthesized and Modified in the Endoplasmic Reticulum
 - 7-4-2. Secretory Proteins Are Modified in the Golgi Complex
- 7-5. Surface Membrane and Extracellular Substances Are Recycled in the Cell
- 7-6. Glial Cells Play Diverse Roles in Neural Function
 - 7-6-1. Glia Form the Insulating Sheaths for Axons
 - 7-6-2. Astrocytes Support Synaptic Signaling
 - 7-6-3. Microglia Have Diverse Functions in Health and Disease
 - 7-6-4. Choroid Plexus and Ependymal Cells Produce Cerebrospinal Fluid

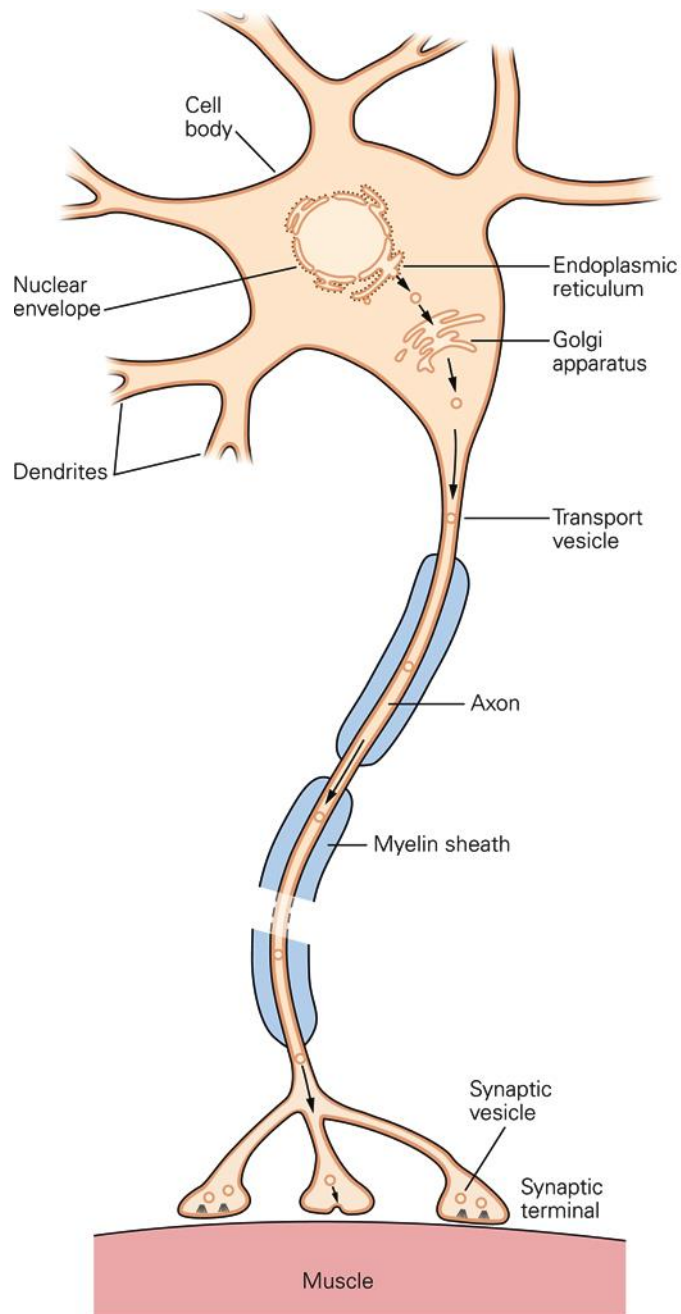
7-1. Neurons and Glia: Shared Origins, Distinct Specializations

Most neurons and macroglia arise from the neuroepithelial lineage.

Microglia are the major exception and have a distinct myeloid origin.

Neurons are highly polarized cells

- **Soma:** cell body containing the nucleus and most biosynthetic organelles.
- **Dendrites:** specialized mainly for receiving and integrating synaptic input.
- **Axon:** specialized for long-distance conduction and presynaptic output.
- Neuronal compartments differ in membrane proteins, cytoskeleton, and organelle distribution.
- **Terminology:** plasmalemma = plasma membrane; axolemma = axonal plasma membrane.

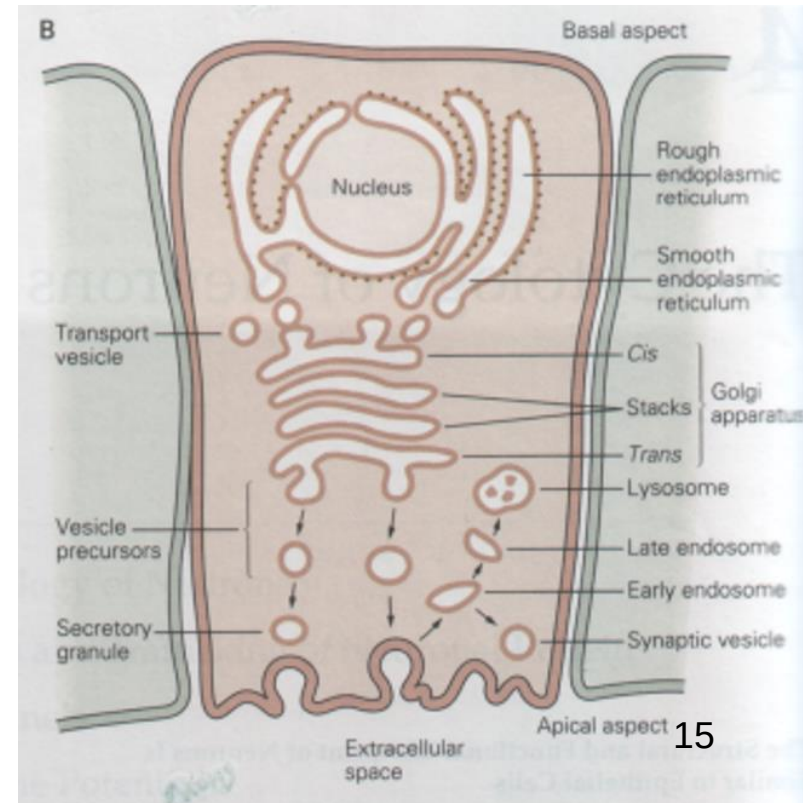


A neuron is a polarized secretory cell.

The nuclear envelope is continuous with the endoplasmic reticulum (ER).

Rough ER and Golgi are concentrated in the soma and can extend into dendrites.

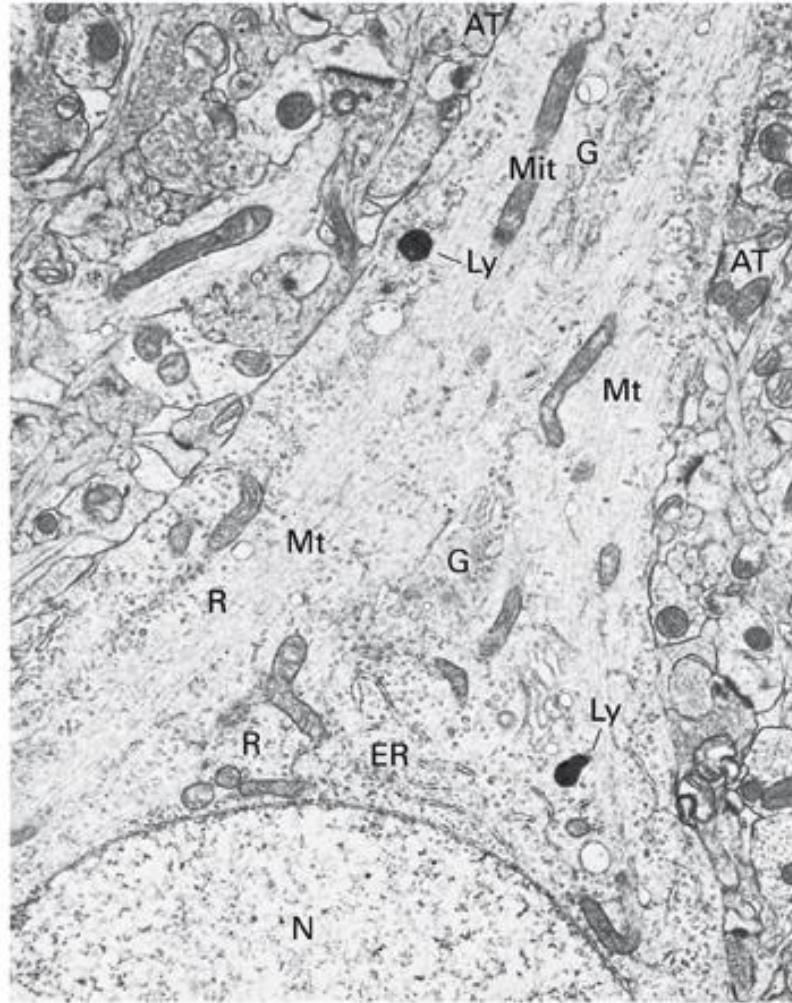
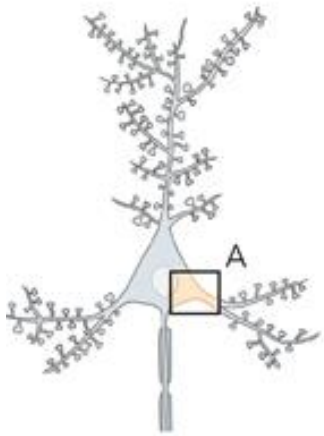
Axons depend strongly on long-distance intracellular transport.



Neuronal organelles are distributed asymmetrically

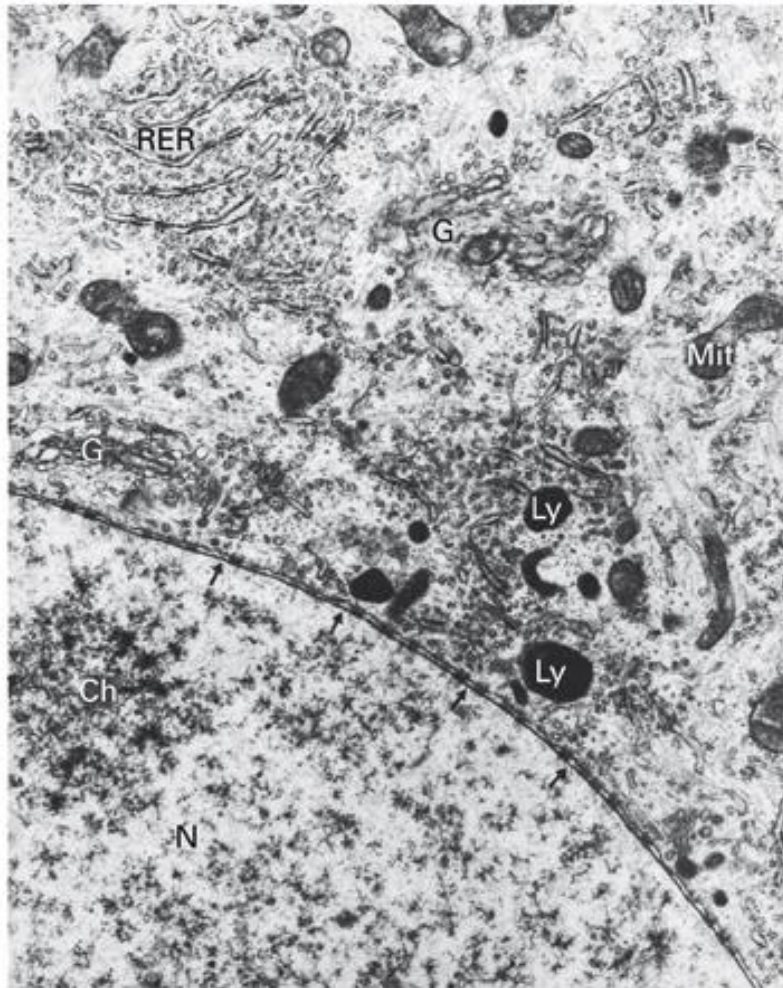
- **Soma and dendrites** contain most of the protein-synthesis machinery, including ribosomes, rough ER, and Golgi.
- **Axons** largely lack rough ER and Golgi and therefore depend strongly on transport from the soma.
- Axons contain abundant transport vesicles, endosomal compartments, mitochondria, and smooth ER.
- Compartment-specific organelle distribution is a key consequence of **neuronal polarity**.

A



Electron micrograph of a dendrite near a pyramidal neuron soma. ER, Golgi, mitochondria, lysosomes, ribosomes, and microtubules are visible.

B



Electron micrograph of a spinal motor neuron soma. Nuclear pores, rough ER, Golgi, lysosomes, and mitochondria illustrate the strong biosynthetic capacity of the cell body.

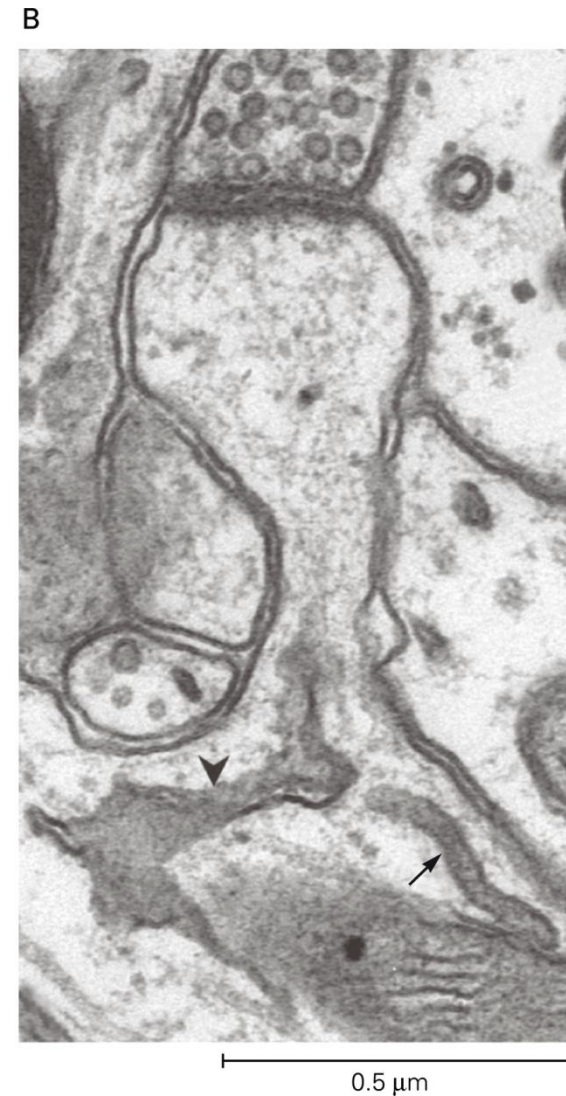
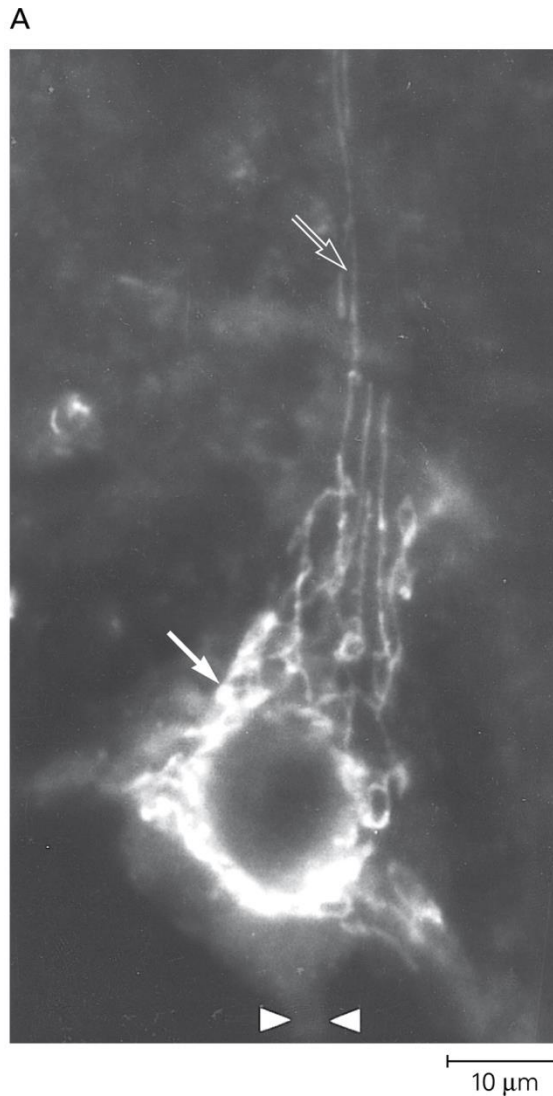
Golgi and rough ER extend into dendrites but are largely excluded from axons

- Dendrites share many biosynthetic organelles with the soma, although their abundance generally decreases with distance.
- Smooth ER is present in dendrites and can extend into dendritic spines.
- Polyribosomes near dendritic spines can support **local protein synthesis** in response to synaptic activity.

A. Golgi membranes can extend from the soma into dendrites but not into the axon.

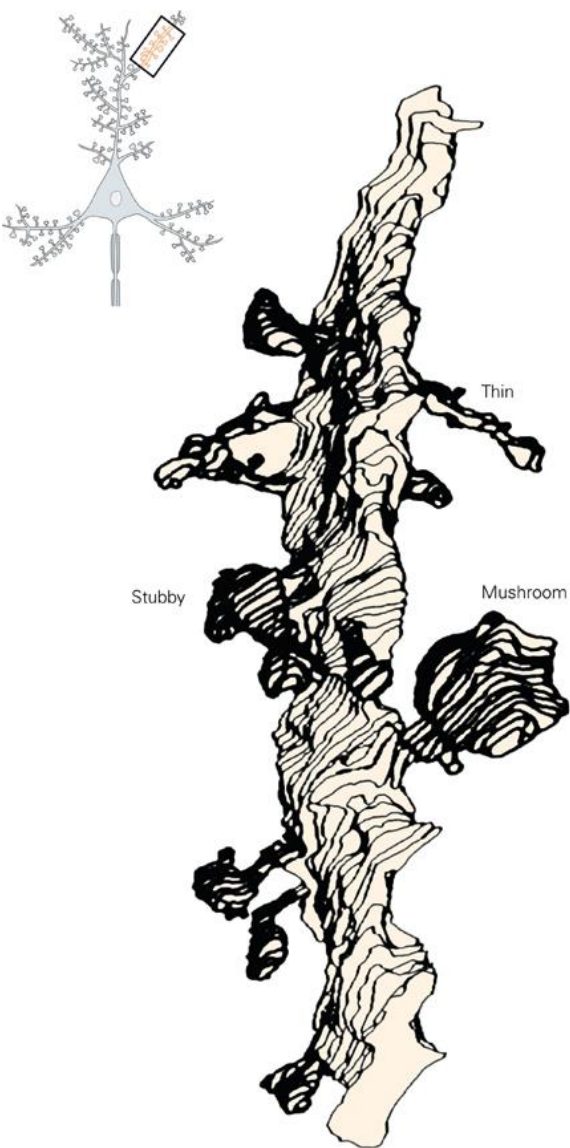
B. Smooth ER can extend into the neck of a dendritic spine.

These images illustrate the compartmentalized organization of neuronal organelles.

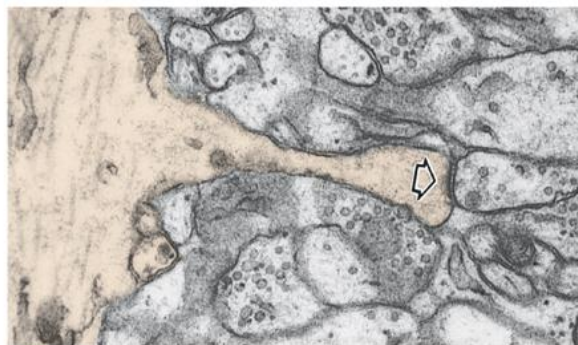


Dendritic spines

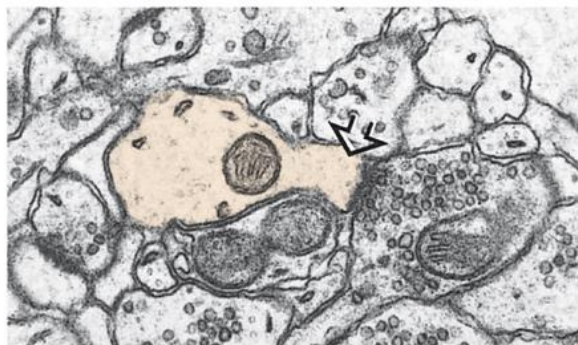
- **Dendritic spines** are small membranous protrusions from dendrites and are common sites of **excitatory synapses**.
- The spine head contains postsynaptic receptors and the postsynaptic density; a thin neck connects it to the dendritic shaft.
- Spine number and structure are **dynamic** and can change with synaptic activity.
- Dendritic spines provide a structural substrate for **synaptic plasticity**.



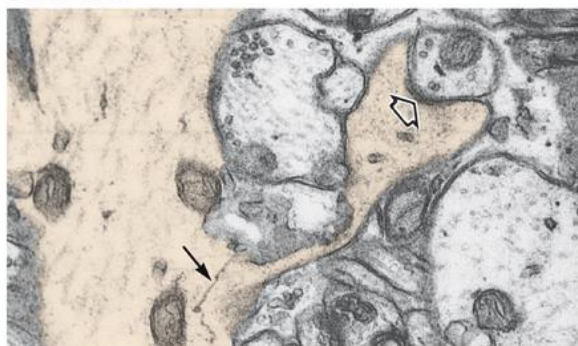
A Thin



B Stubby



C Mushroom

10 μ m

Thin, stubby, and mushroom are descriptive spine morphologies.

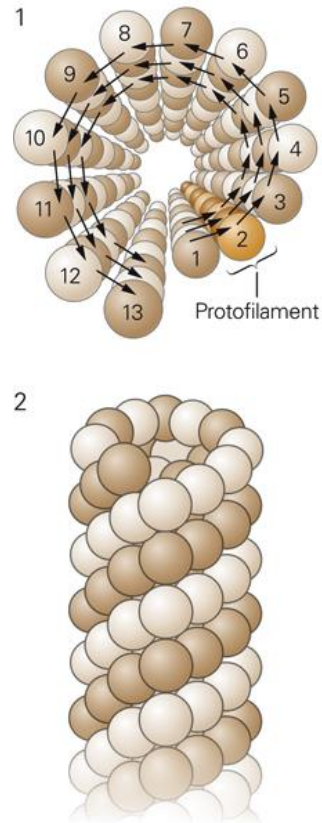
Spine morphology is dynamic and relates to synaptic organization.

The electron micrographs illustrate a postsynaptic density and the close apposition of pre- and postsynaptic elements.

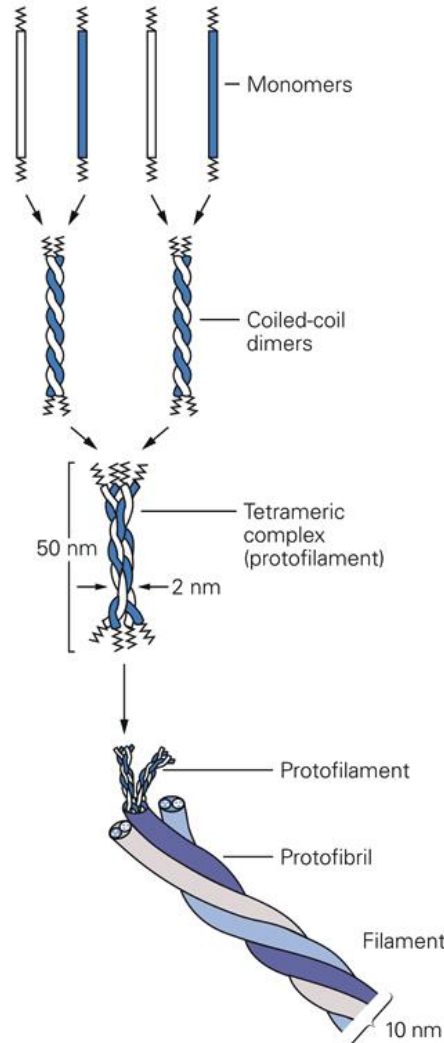
7-2. The Cytoskeleton Determines Cell Shape

Three major cytoskeletal systems in neurons

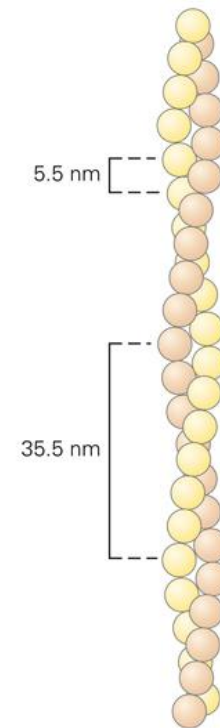
A Microtubule

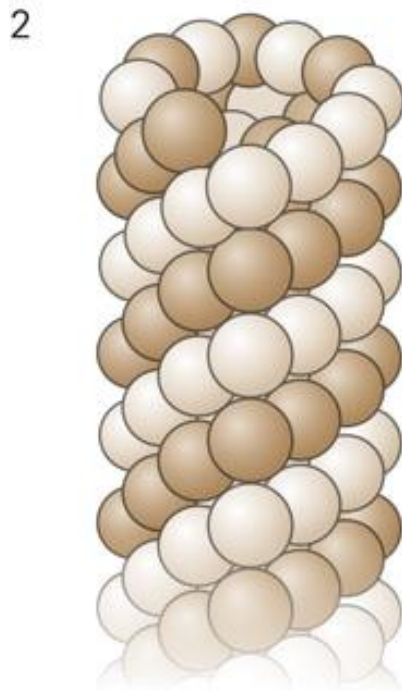
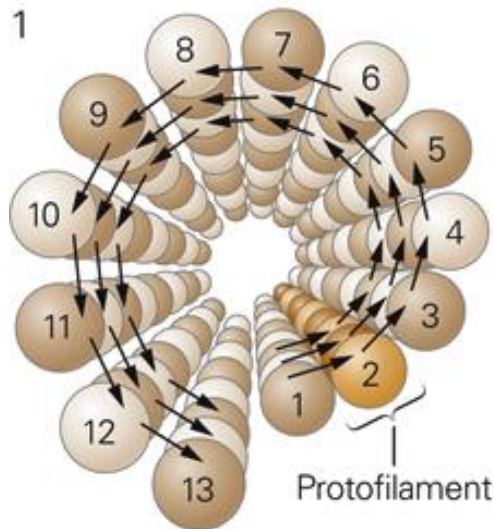


B Neurofilament



C Microfilament





- **Microtubules** are hollow polymers of α/β -tubulin heterodimers (~25 nm diameter).
- They are **polar** structures with distinct + and – ends.
- Microtubule polarity is essential for directional intracellular transport in axons and dendrites.

The diagram shows the detailed lattice architecture.

1. Microtubules

- Microtubules are dynamic polymers built from α/β -tubulin and use GTP.
- The **+ end** usually grows and shrinks more rapidly than the – end.
- Their structural polarity provides directional tracks for molecular motors.
- In mature neurons, microtubules are relatively stabilized but remain dynamic.

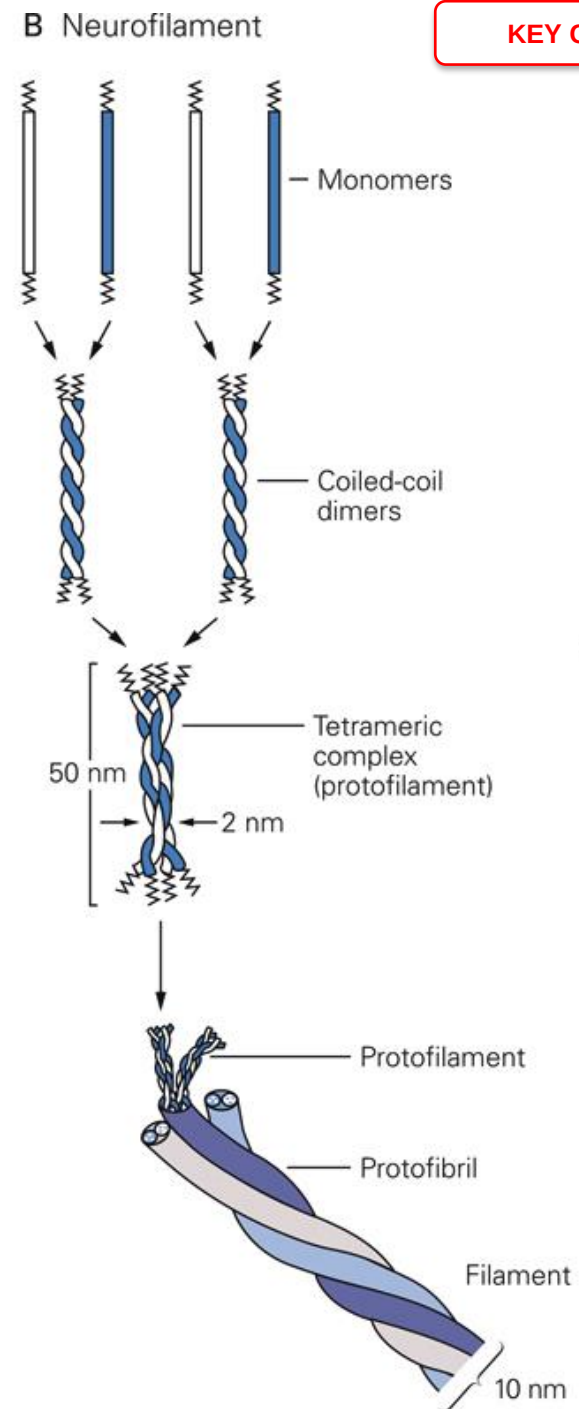
TABLE 20-1 Proteins That Modulate Microtubule (MT) Dynamics

Protein	MW	Location	Function
MICROTUBULE-STABILIZING PROTEINS			
MAP1	250,000–300,000 (heavy chain)	Dendrites and axons; non-neuronal cells	Assembles and stabilizes MTs
MAP2	42,000 and 200,000	Dendrites	Assembles and cross-links MTs to one another and to intermediate filaments
MAP4	210,000	Most cell types	Stabilizes MTs
Tau	55,000–62,000	Dendrites and axons	Assembles, stabilizes, and cross-links MTs
CLIP170	170,000	Most cell types	Cross-links MTs to endosomes and chromosomes
MICROTUBULE-DESTABILIZING PROTEINS			
Katanin	84,000	Most cell types	Microtubule severing
Op18 (stathmin)	18,000	Most cell types	Binds tubulin dimers

Selected microtubule-associated proteins (MAPs), which regulate microtubule stability and organization.

2. Neurofilaments

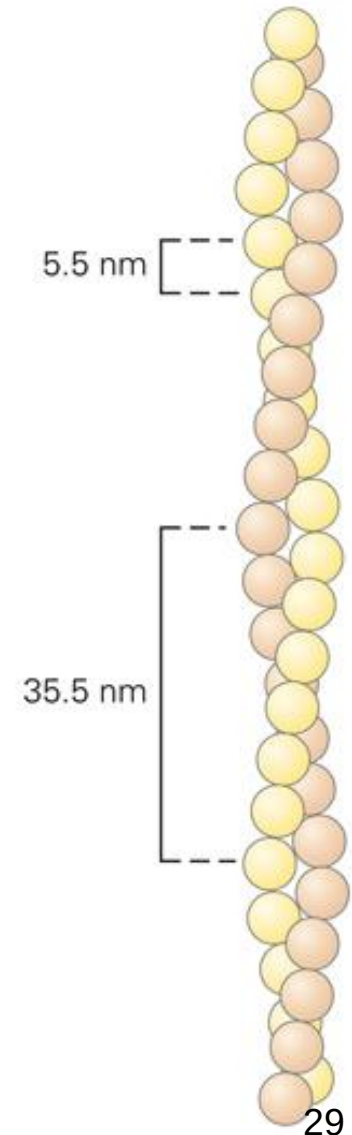
- Neurofilaments are neuron-enriched **intermediate filaments** (~10 nm).
- They are relatively stable polymers that provide structural support, especially in axons.
- Neurofilament content helps determine **axonal caliber** and mechanical strength.
- Composed mainly of NF-L (light), NF-M (medium), and NF-H (heavy), together with α -internexin in the CNS or peripherin in the PNS.



3. Microfilaments

- **Actin microfilaments** (~7 nm) are polymers of globular actin.
- They are **polar and highly dynamic** structures whose assembly is ATP-dependent.
- Actin is enriched beneath the plasma membrane and in growth cones, presynaptic terminals, and dendritic spines.
- Actin remodeling is essential for changes in neuronal shape and synaptic structure.

C Microfilament



- Microtubules and actin are dynamic systems that support intracellular transport and structural plasticity.
- Neurofilaments provide relatively stable mechanical support, especially in axons.

System	Main subunit	Diameter	Dynamics	Major neuronal role
Microtubule	α/β -tubulin	~25 nm	Dynamic; polarized	Long-distance transport; organization of axons and dendrites
Neurofilament	NF subunits	~10 nm	Relatively stable	Axonal caliber and mechanical support
Microfilament	Actin	~7 nm	Highly dynamic; polarized	Cell shape, growth cones, presynaptic terminals, and dendritic spines

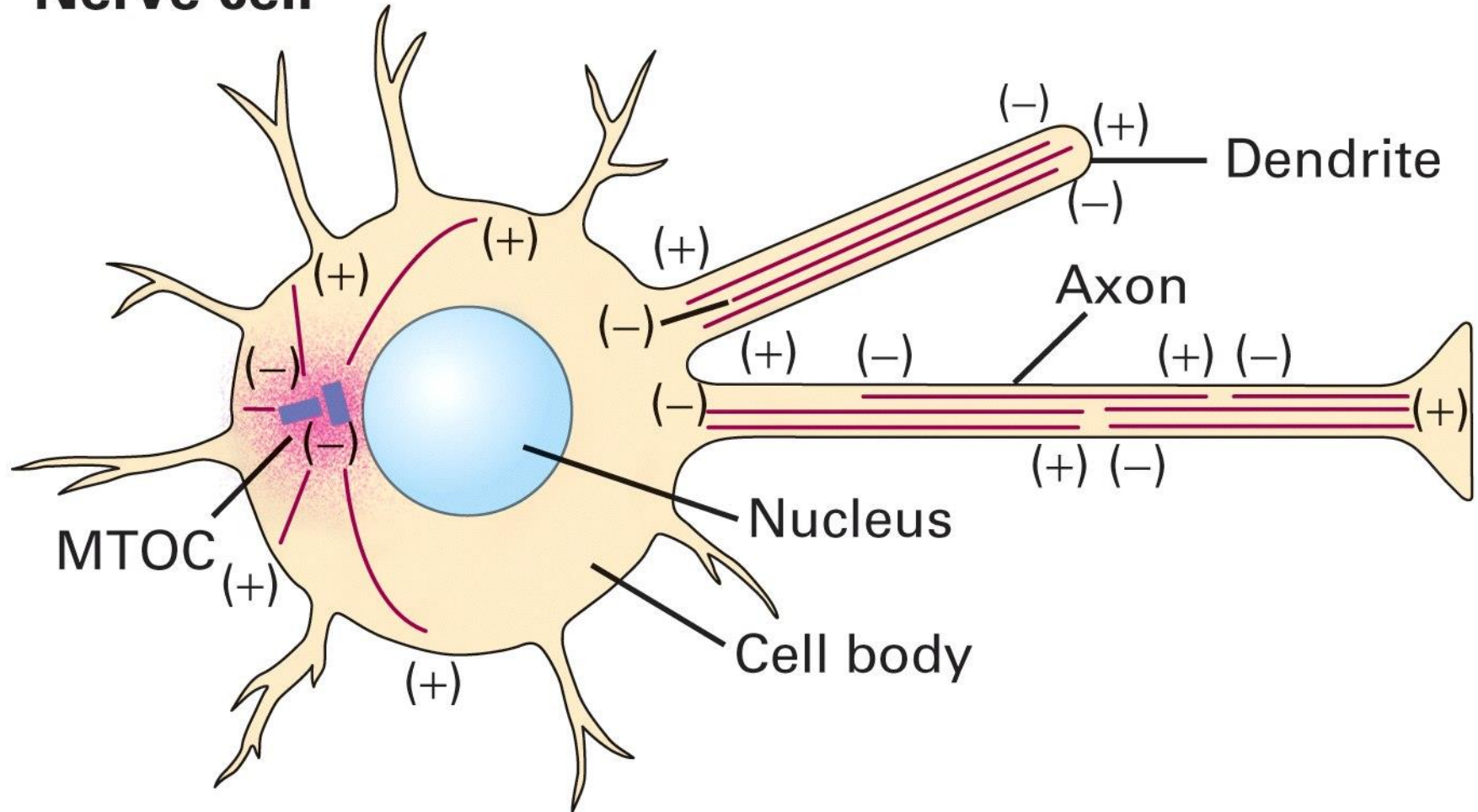
7-3. Axonal and Dendritic Transport

- Neurons must move **vesicles, organelles, proteins, and RNAs** over long distances.
- **Anterograde transport**: cell body → axon terminal; supports axonal growth and synaptic supply.
- **Retrograde transport**: nerve terminal → cell body; returns materials and carries signals.
- **Axonal microtubules** usually have their “+” ends toward the terminal; dendritic microtubules have mixed polarity.

Microtubule polarity gives direction to transport

KEY CONCEPT

Nerve cell



MTOC: microtubule-organizing center
(+)/(-): microtubule polarity

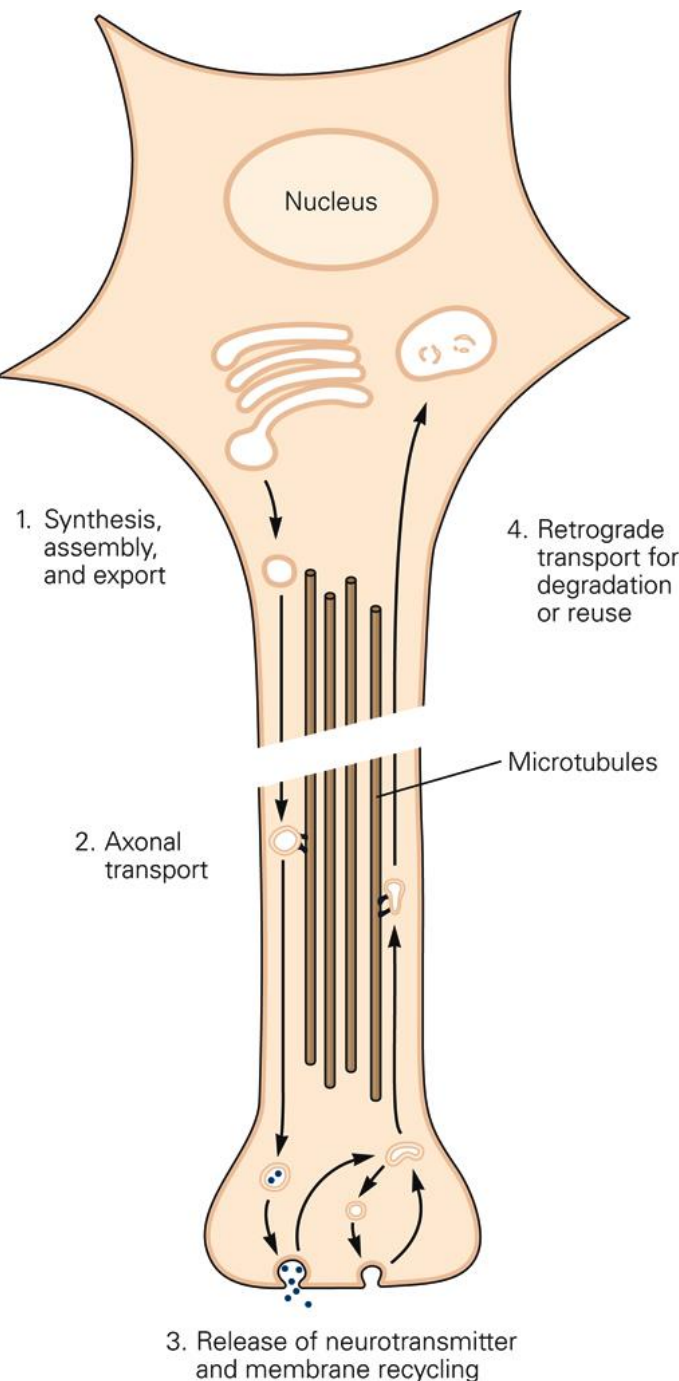


Figure 7-9: Membrane trafficking in neurons

Secretory/membrane proteins are synthesized in ER and processed through Golgi.

Transport vesicles move cargo down the axon.

Synaptic vesicles release neurotransmitters and are locally recycled.

Dense-core vesicles carry peptide transmitters.

Pulse-labeling experiments revealed that newly synthesized proteins move along axons.

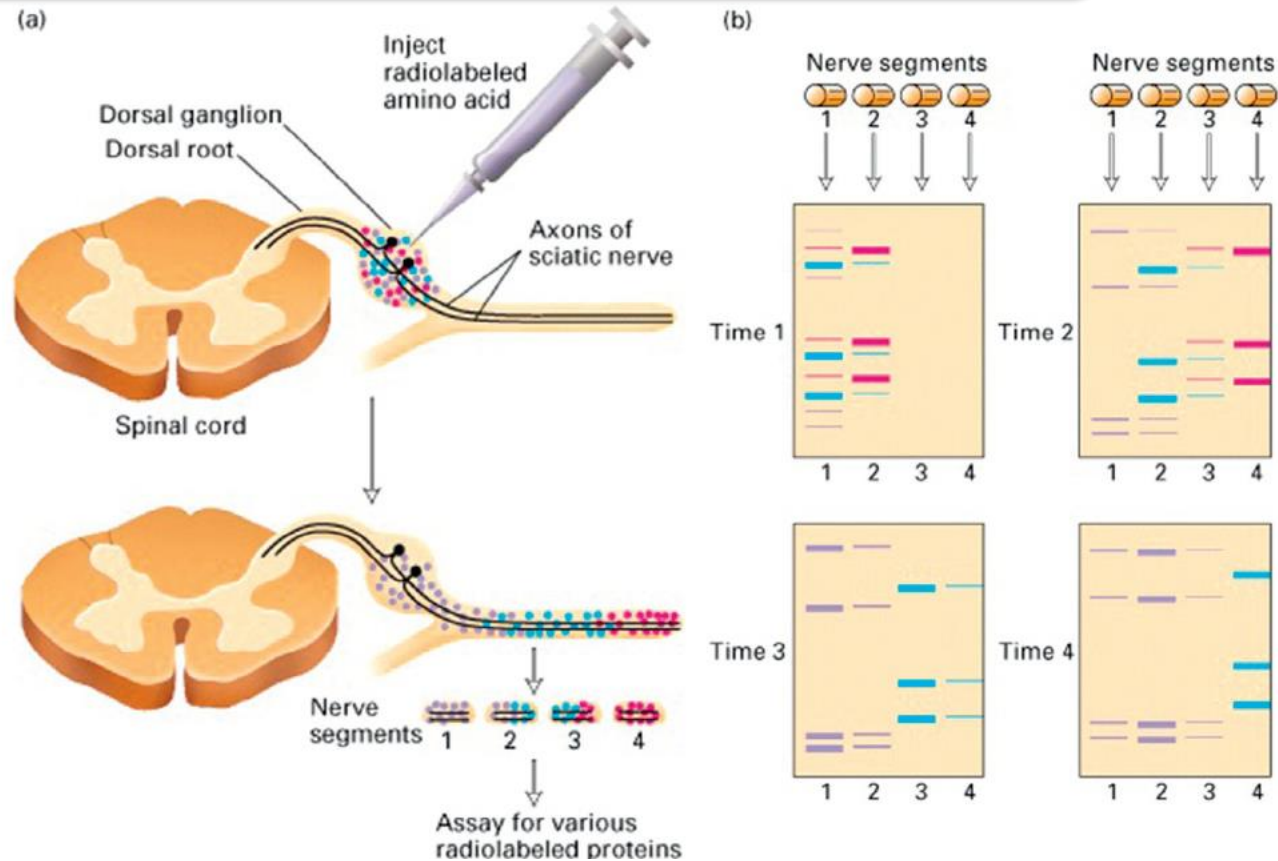
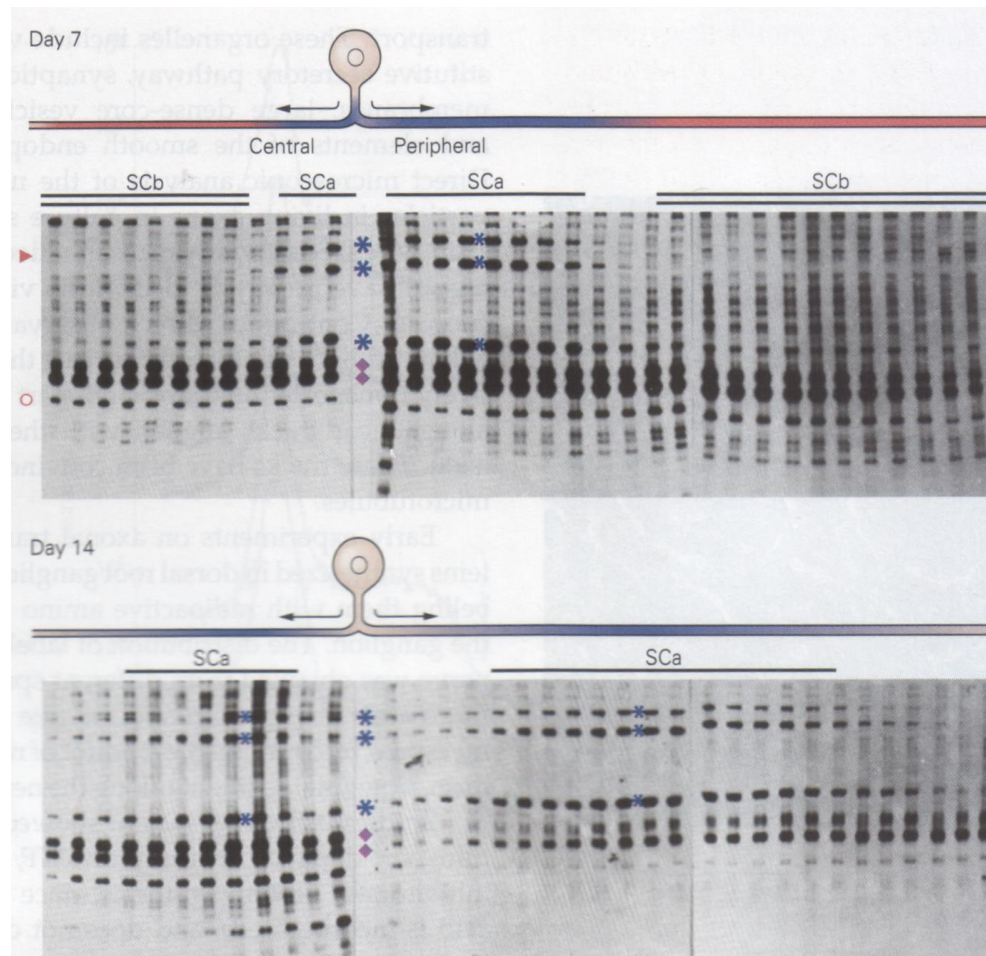
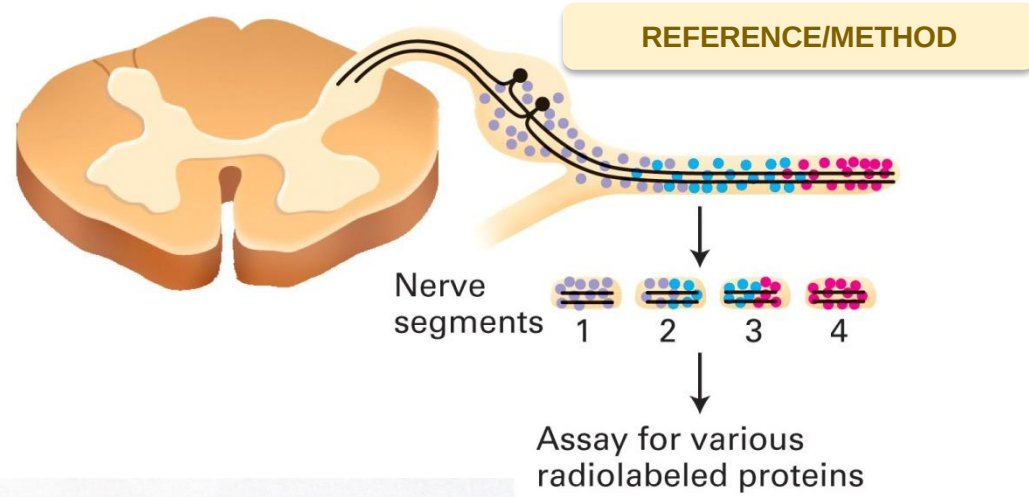
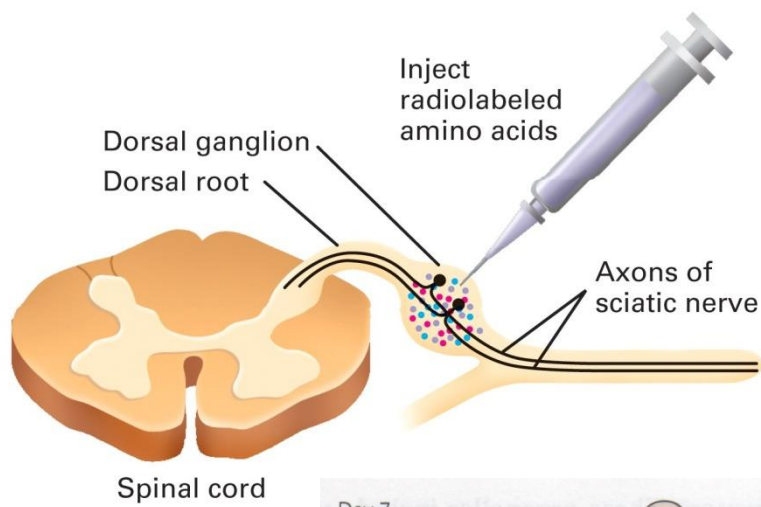


Figure 2 The radioisotopic pulse-labeling technique. (a) Radiolabeled amino acids are injected into the vicinity of nerve cell bodies (a dorsal root ganglion in this example), and the amino acids are incorporated into proteins in the nerve cell bodies. The proteins move out along the axons in association with distinct cargo structures (shown here as purple, blue, and red dots), which move at different rates. At early times the faster- and slower-moving cargo structures overlap considerably, whereas at later times they become separated spatially. At various time intervals, the animals are sacrificed and the nerves are removed and cut into contiguous segments. (b) For each time interval (time 1–4), the proteins in each nerve segment are separated by gel electrophoresis and the amount of radioactivity associated with each protein is quantified. Labeled proteins associated with the purple, blue, and red cargo structures in (a) are shown as purple, blue, and red bands, respectively, in (b). Reproduced from Lodish H, Berk A, Zipursky SL, Matsudaira P, Baltimore D, and Darnell J (2000) *Molecular Cell Biology*, 4th edn. New York: W. H. Freeman.



Fast and slow axonal transport

- **Fast transport:** membranous cargo, such as vesicles, endosomes, mitochondria, and smooth ER.
- **Direction:** can be anterograde or retrograde, depending on the motor protein.
- **Slow transport:** mainly cytosolic and cytoskeletal proteins moving from soma into axon.
- **Key distinction:** cargo type and speed, not simply whether movement is forward or backward.

Fast axonal transport carries membranous cargo

- **Typical cargo:** synaptic vesicle precursors, dense-core vesicles, mitochondria, smooth ER, endosomes, and RNA-containing particles.
- **Track:** microtubules aligned along the axon.
- **Energy:** ATP-dependent movement driven by molecular motors.
- **Movement pattern:** cargo can move in a stop-and-start (saltatory) fashion along the microtubule track.

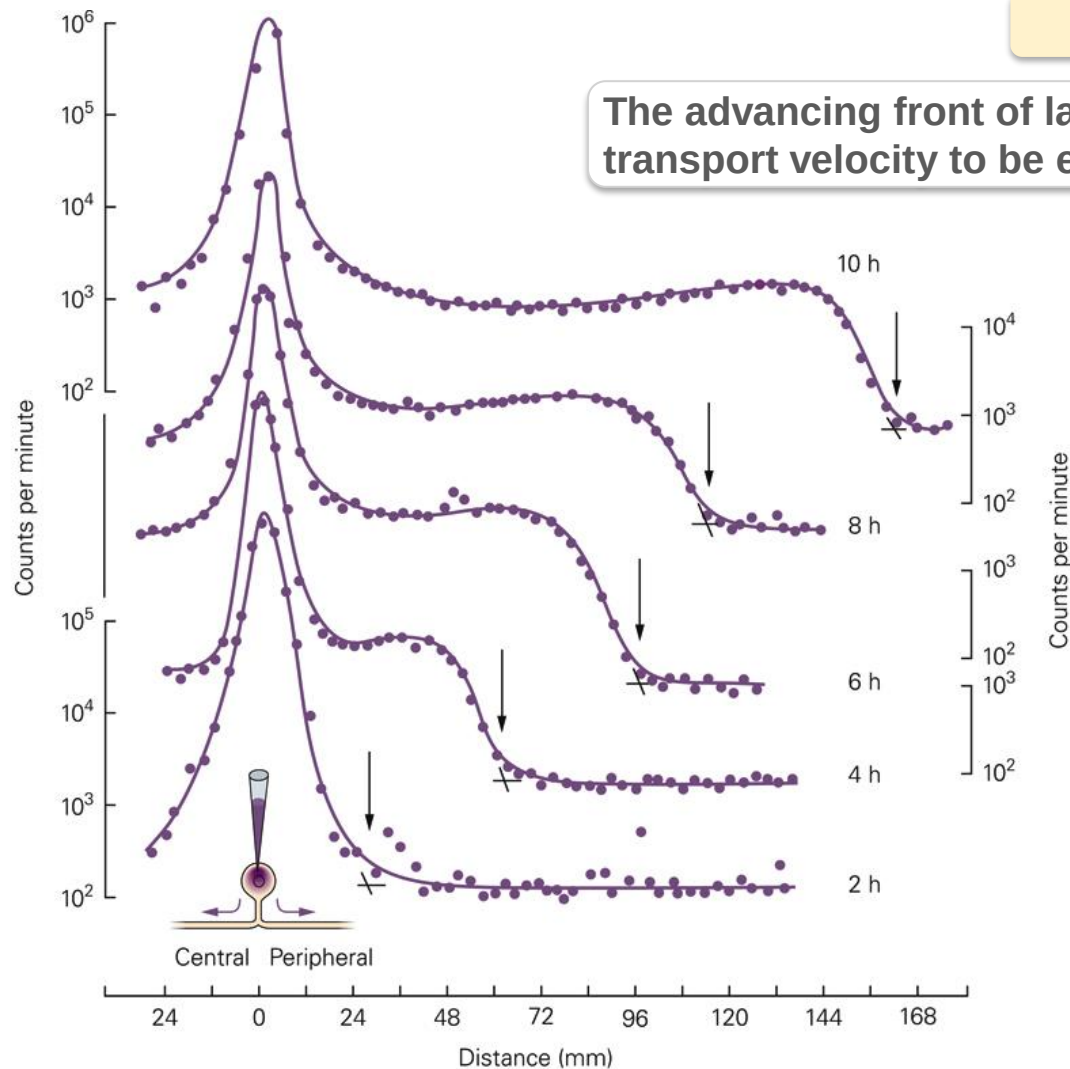


Figure 7-11 Early experiments on anterograde axonal transport used radioactive labeling of proteins. In the experiment illustrated here the distribution of radioactive proteins along the sciatic nerve of the cat was measured at various times after injection of $[^3\text{H}]$ -leucine into dorsal root ganglia in the lumbar region of the spinal cord. To show transport curves from various times (2, 4, 6, 8, and 10 hours after the injection) in one figure, several ordinate scales (in logarithmic units) are used. Large amounts of labeled protein stay in the ganglion cell bodies but,

with time, move out along axons in the sciatic nerve, so the advancing front of the labeled protein is progressively farther from the cell body (**black arrows**). The velocity of transport can be calculated from the distances of the front displayed at the various times. From experiments of this kind Sidney Ochs found that the rate of fast axonal transport is constant at 410 mm per day at body temperature. (Adapted, with permission, from Ochs 1972.)

Some mRNAs and ribosomes are transported into dendrites.

Dendritic local translation allows selected proteins to be synthesized near synapses.

Functional idea: local protein synthesis helps support activity-dependent synaptic plasticity.

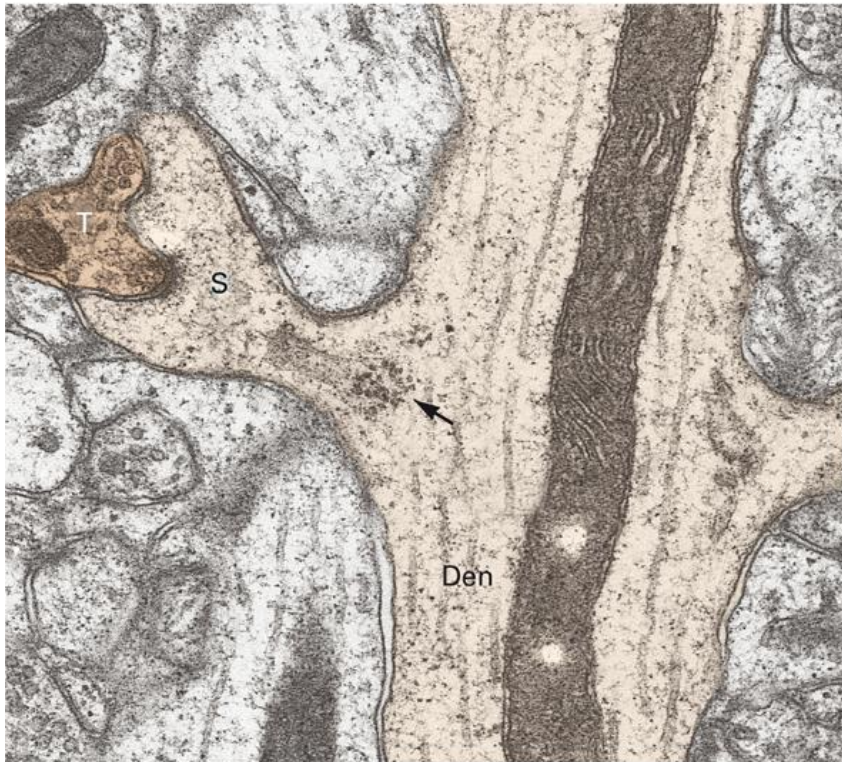


Figure 7-12. Ribosomes in the dendritic arbors.

Ribosomes in dendrites are selectively concentrated at the junction of the spine and main dendritic shaft (arrow), where the spine contacts the axon terminal of a presynaptic neuron.

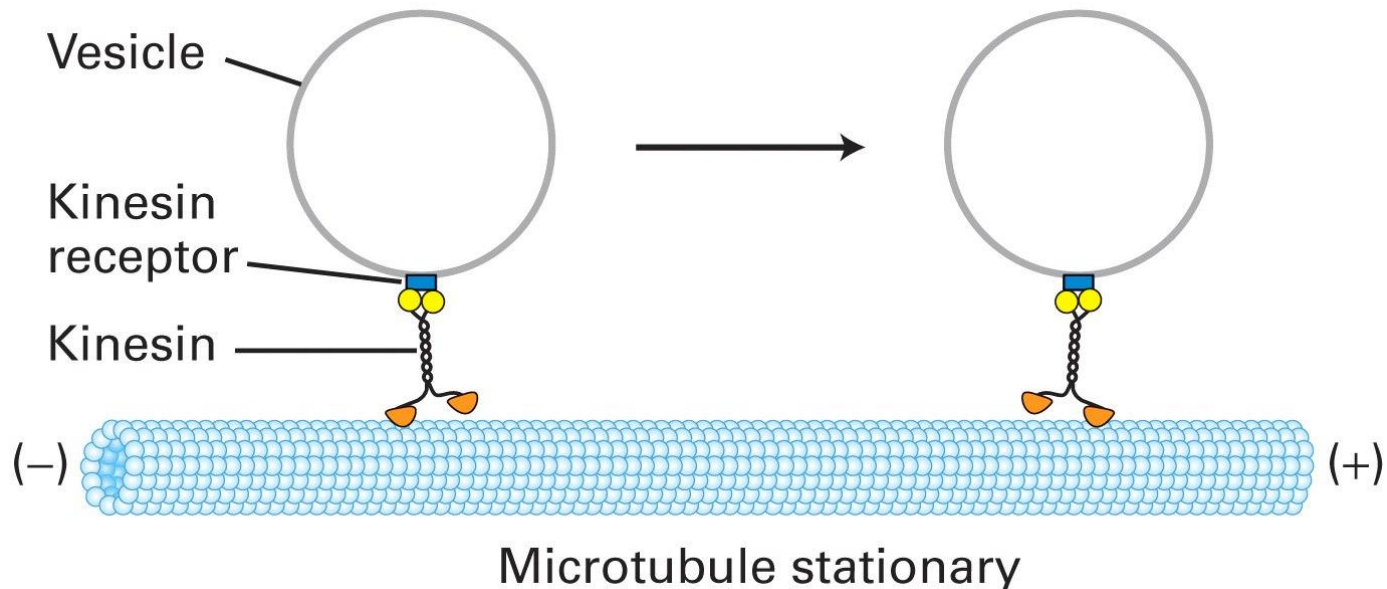
This electron micrograph shows a mushroom-shaped spine of a neuron in the hippocampal dentate gyrus. Note the absence of ribosomes in the dendritic shaft. S, spine head; T, presynaptic terminal; Den, main shaft of the dendrite containing a long mitochondrion. × 60,000.

Molecular motors convert ATP into directed movement

- **Cargo does not simply diffuse** along long axons at biologically useful rates.
- **Motor proteins** bind cargo and walk along microtubule tracks.
- **Kinesins** are ATP-driven motors; many move toward microtubule “+” ends.
- **Different motors** carry different cargoes in different neuronal compartments.

Kinesin: the major anterograde motor

- **Kinesin** generally moves cargo toward the microtubule “+” end.
- **In axons**, the microtubule “+” end usually points toward the terminal; kinesin therefore supports anterograde transport.

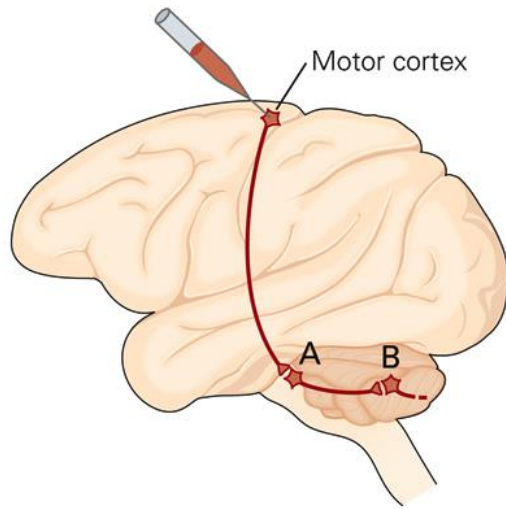


Retrograde fast transport returns cargo and signals

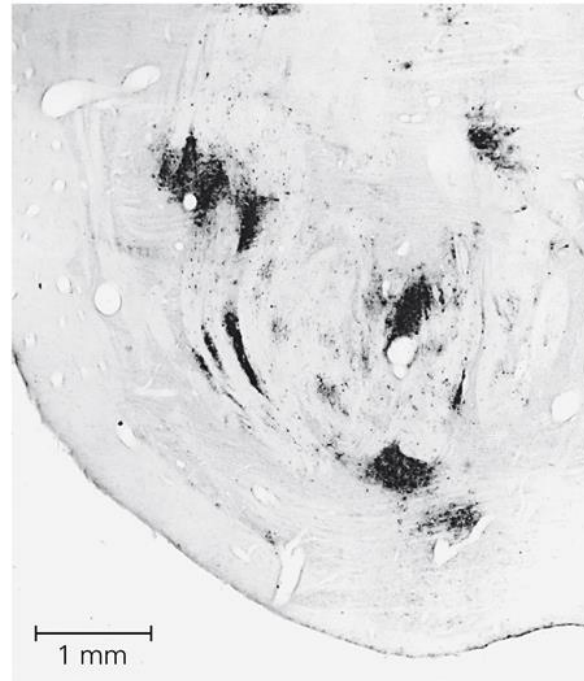
- **Retrograde cargo:** endosomes, mitochondria, ER-derived components, and damaged materials.
- **Signaling function:** activated receptors (such as neurotrophic receptors) and signaling complexes can travel from terminals back to the nucleus.
- **Clinical/biological examples:** some toxins and viruses exploit retrograde axonal transport.

HSV tracing illustrates how pathogens can move along connected neuronal pathways.

REFERENCE/EXAMPLE



A Pontine nuclei



B Cerebellar cortex

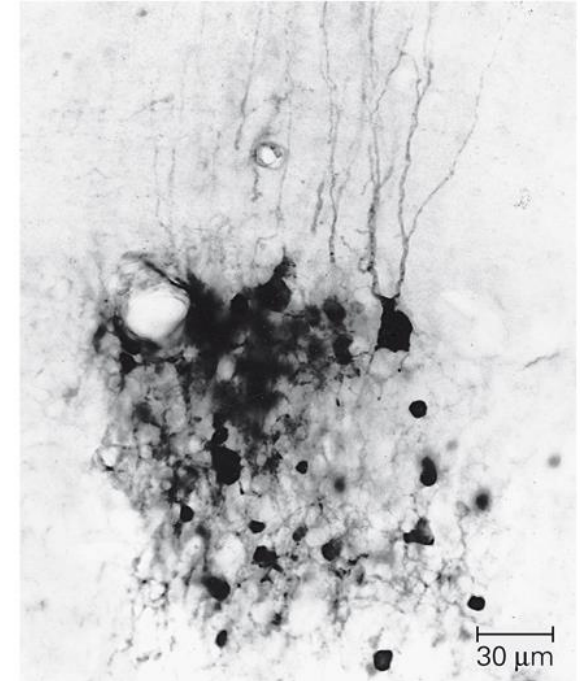


Fig. 7-10 Axonal transport of the herpes simplex virus (HSV) is used to trace cortical pathways in monkeys. Depending on the strain, the virus moves in the anterograde or retrograde direction by axonal transport. In either direction it enters a neuron with which the infected cell makes synaptic contact. Here the projections of cells in the primary motor cortex to the cerebellum in monkeys were traced using an anterograde-moving strain (HSV-1

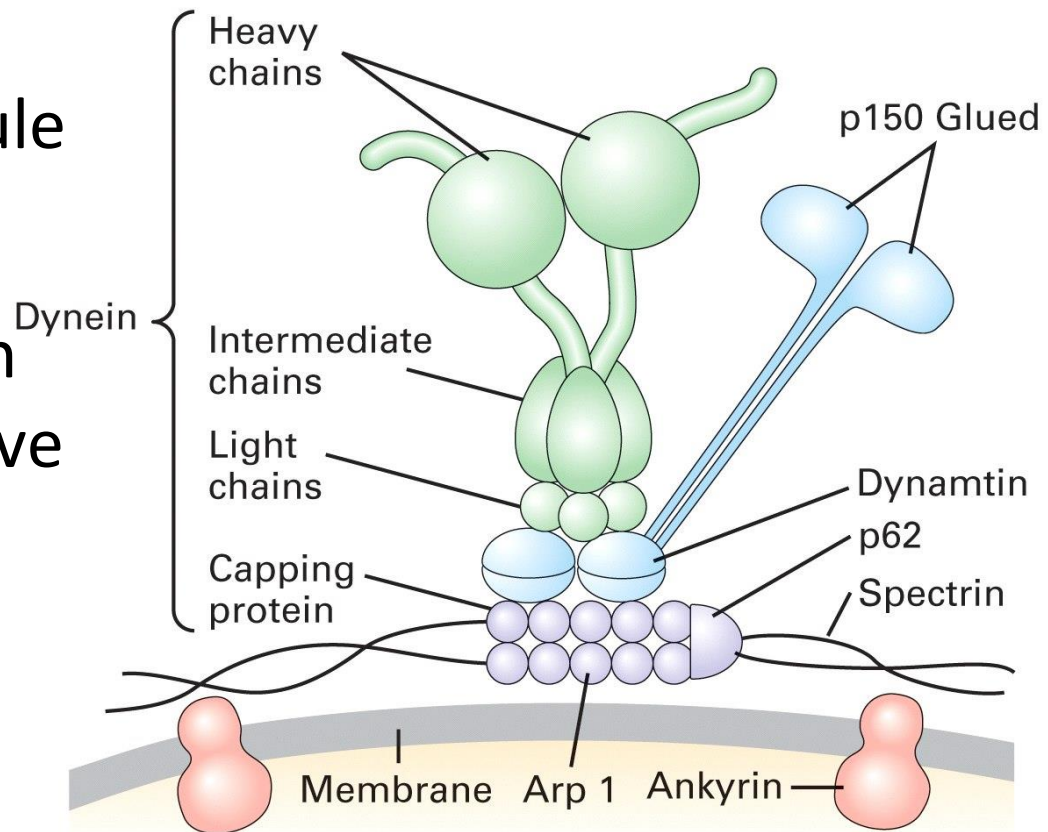
[H129]). Monkeys were injected in the region of the primary motor cortex that controls the arm. After 4 days the brain was sectioned and immunostained for viral antigen. Micrographs show the virus was transported from the primary motor cortex to second-order neurons in pontine nuclei (A) and then to third-order neurons in the cerebellar cortex (B). (Reproduced, with permission, from P. L. Strick.)

Dynein and dynactin mediate retrograde transport

- **Cytoplasmic dynein** is the main motor for movement toward the microtubule “–” end.
- **In axons**, “–” end-directed movement generally carries cargo back toward the cell body.
- **Dynactin** helps link dynein to cargo and improves dynein-based transport.
- **Just** remember dynein = retrograde/“–” end.

Dynein: the major retrograde motor

- **Dynein** moves cargo toward the microtubule “-” end.
- **Dynactin** helps dynein engage cargo and move efficiently.



Slow axonal transport maintains axonal structure

- **Slow transport** mainly moves cytosolic and **cytoskeletal** proteins from the soma into the axon.
- **It is much slower** than vesicular fast transport and is predominantly anterograde.
- **Typical cargo**: neurofilament proteins, α/β -tubulin, actin, actin-binding proteins, and enzymes.
- **Function**: renew long-lived axonal components and maintain axonal caliber and organization.

7-4. Protein Synthesis and Membrane Trafficking in Neurons

Secretory and membrane proteins are made on rough ER

- mRNAs encoding secreted or membrane proteins are translated by ribosomes attached to the rough ER.
- During translation, the growing polypeptide enters the ER lumen or becomes inserted into the ER membrane.
- The ER is the first major station for folding and early modification before proteins move to the Golgi.
- ER: endoplasmic reticulum

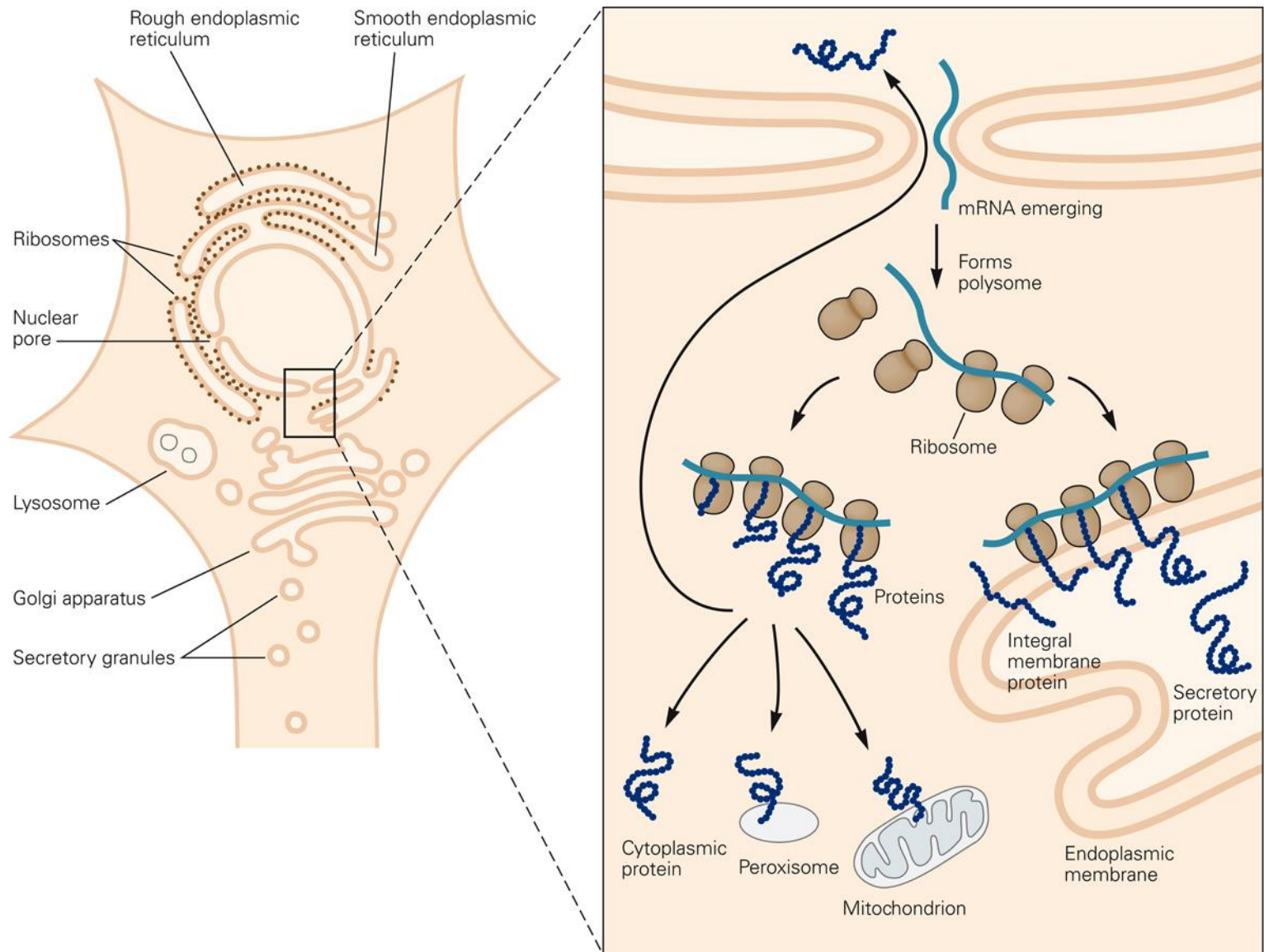


Figure 7-13 Protein synthesis in the endoplasmic reticulum. Free and membrane-bound polysomes translate mRNA that encodes proteins with a variety of destinations. Messenger RNA, transcribed from genomic DNA in the neuron's nucleus,

emerges into the cytoplasm through nuclear pores (enlarge-ment) to form polyribosomes. The polypeptides that become secretory and membrane proteins are translocated across the membrane of the rough endoplasmic reticulum.

Two outcomes of ER-associated translation

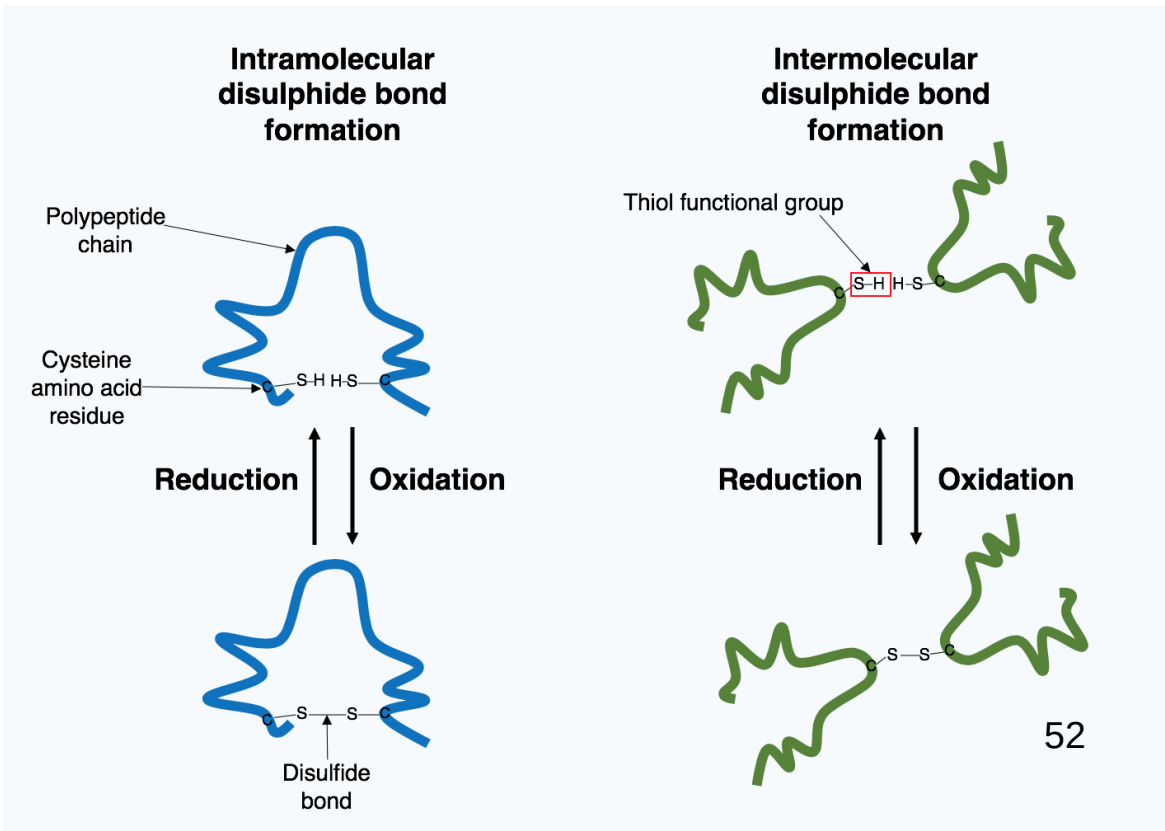
- Soluble proteins **fully enter** the ER lumen and may later be secreted, such as neuropeptides.
- Membrane proteins are inserted into the ER membrane and later become receptors, ion channels, or transporters.
- After ER processing, proteins move to **the Golgi** for further modification and sorting.

Important protein modifications

- After translation, proteins are often chemically modified before they become functional.
- For this course, focus on the following selected concepts:
 - disulfide bond: stabilization of protein structures
 - lipid modification: help proteins to associate with membranes
 - phosphorylation: reversible regulation of protein activity
 - ubiquitination: protein turnover and signaling
 - glycosylation: protein maturation, trafficking, and cell recognition

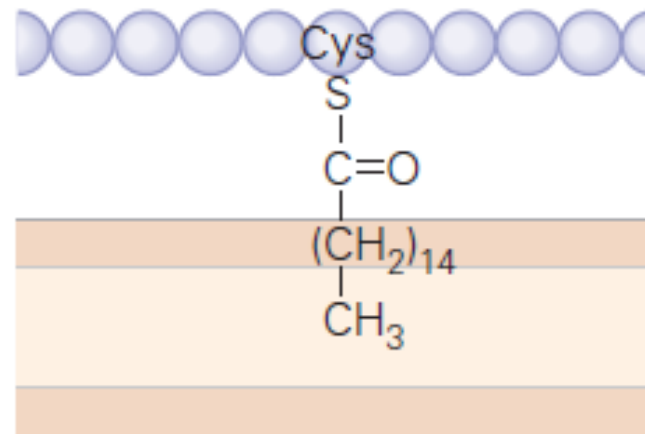
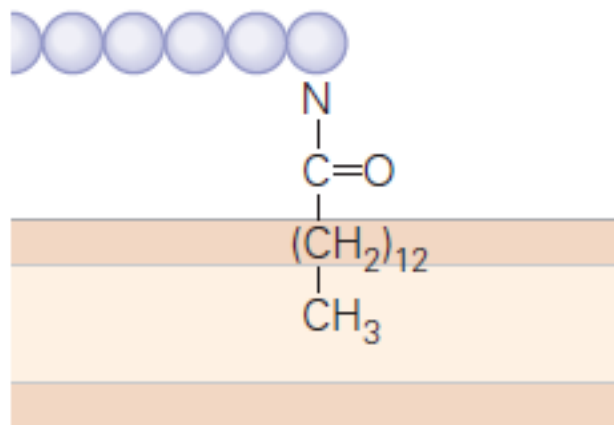
Disulfide bond formation

- Disulfide bonds form between **cysteine** residues in the oxidizing environment of the ER.
- They help stabilize the folded structure of many secreted and membrane proteins.



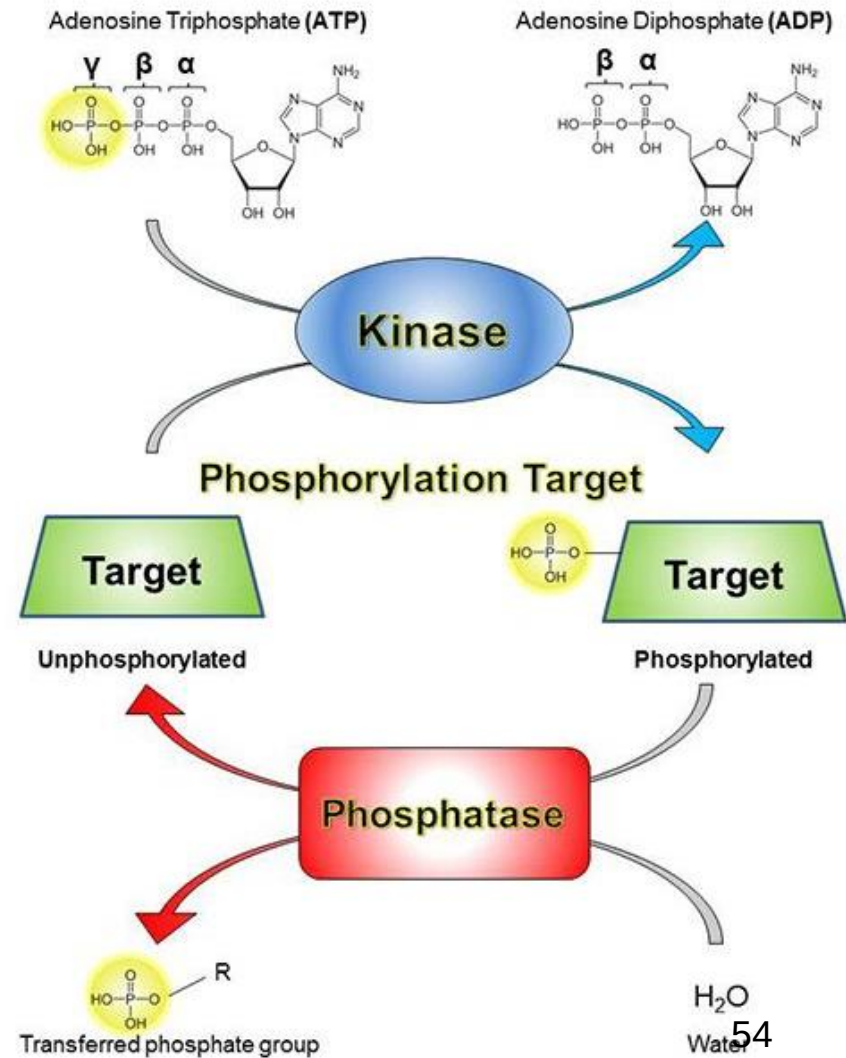
Lipid modifications of proteins

- Some proteins acquire lipid groups during or after translation.
- Lipid attachment can help proteins associate with cellular membranes.
- N-acylation (left) and thioacylation (right) are examples; the chemical mechanisms are not required here.



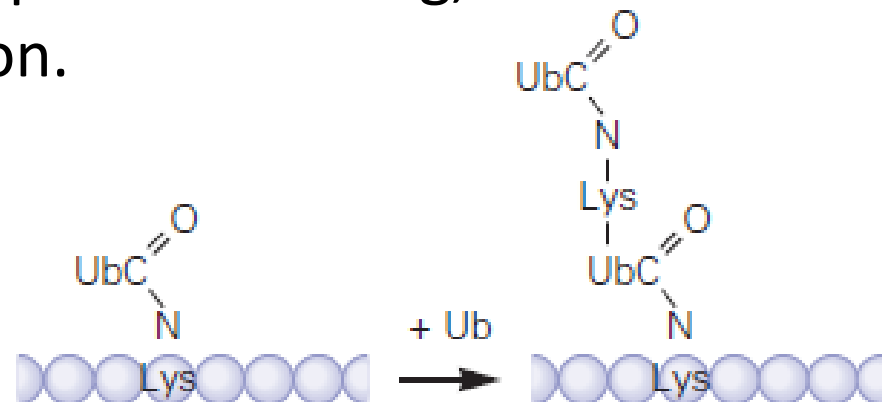
Phosphorylation/dephosphorylation: reversible regulation

- Protein **kinases** add phosphate groups; protein **phosphatases** remove them.
- This modification is **reversible** and can rapidly change protein function.
- Phosphorylation occurs at the hydroxyl group in serine (S), threonine (T), or tyrosine (Y) residues
- In neurons, phosphorylation can regulate ion channels, transcription factors, cytoskeleton, and synaptic signaling.



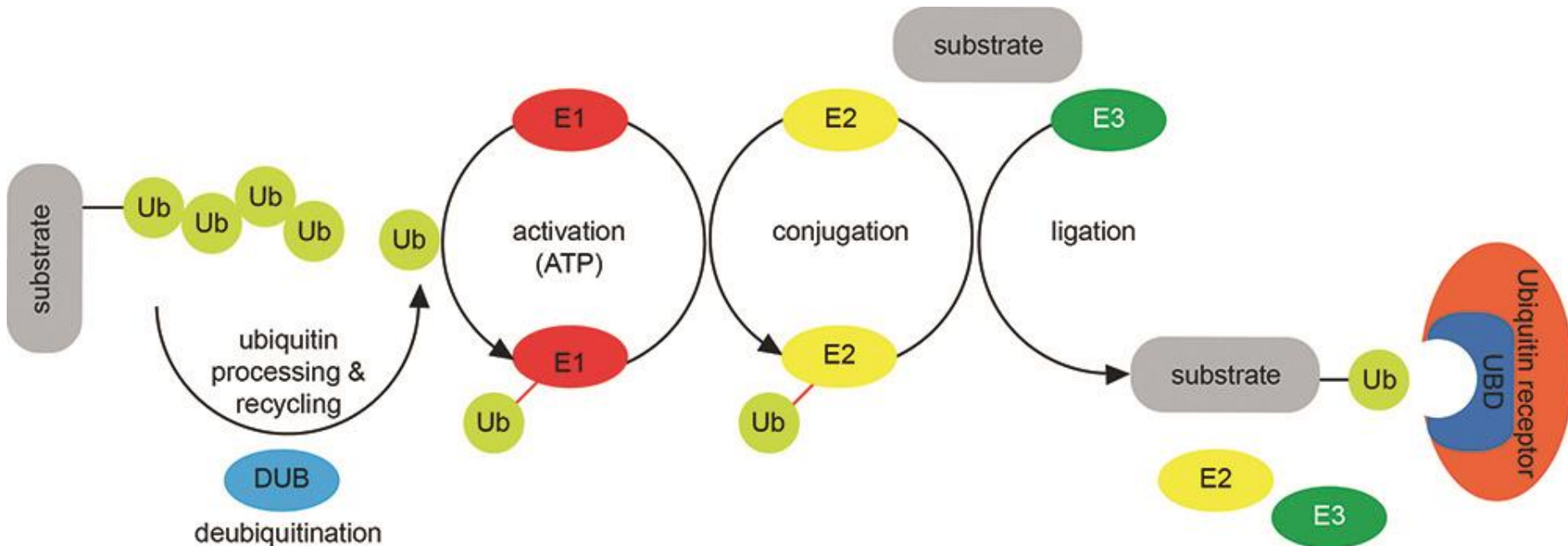
Ubiquitination: tagging proteins for regulation or degradation

- Ubiquitin is a small protein that can be attached to selected target proteins.
- Polyubiquitin chains often mark proteins for degradation by the proteasome.
- Neurons use the ubiquitin–proteasome system for protein quality control, synaptic remodeling, and activity-dependent regulation.



Enzyme cascade for ubiquitination

- Ubiquitination is carried out by an enzyme cascade: $E1 \rightarrow E2 \rightarrow E3$.
- The E3 ligase is especially important because it helps select the target protein.



Glycosylation: sugar modifications of proteins

- Glycosylation adds **carbohydrate chains** to proteins, mainly in the ER and Golgi.
- It is common in membrane and secreted proteins.
- Glycosylation can influence protein folding, trafficking, stability, and cell–cell recognition.
- You do not need to memorize detailed N-linked or O-linked chemistry in this lecture.

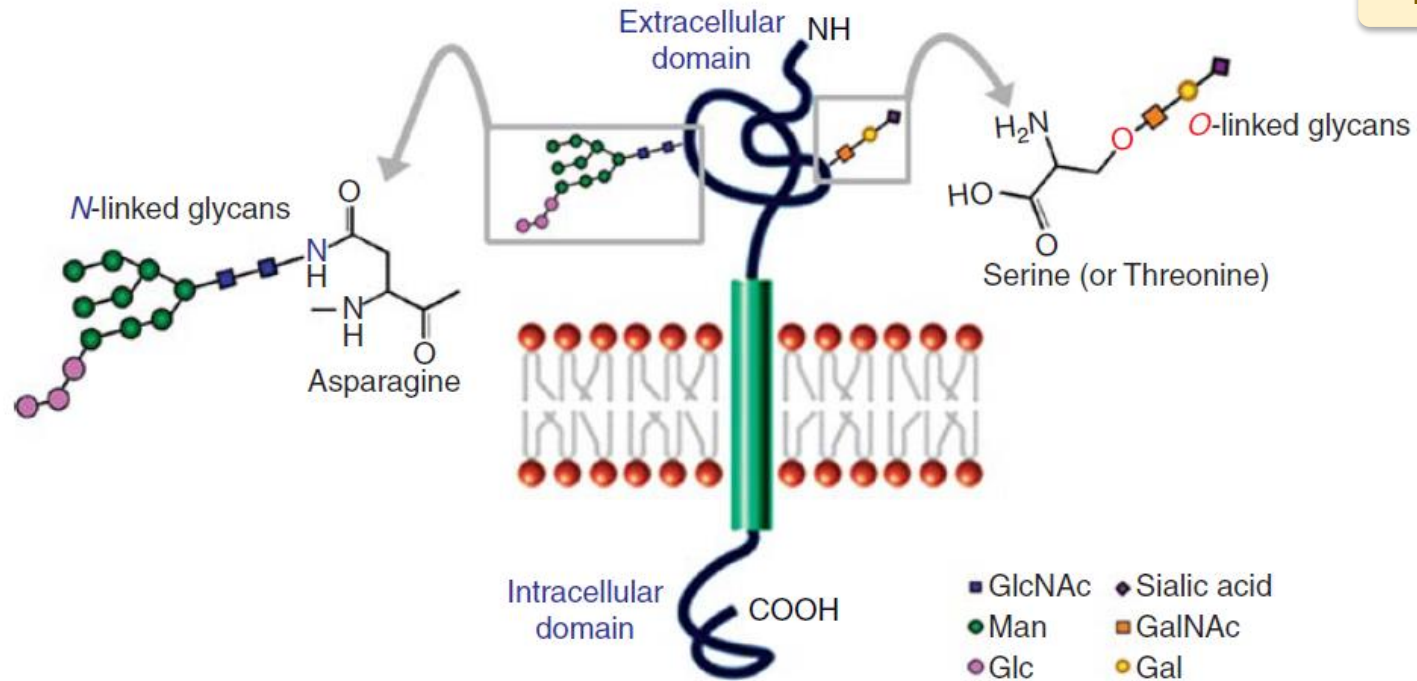


Figure 1. N-linked and O-linked glycosylation on plasma membrane-bound proteins. The glycosylation of plasma membrane-bound proteins is frequently found on the extracellular domains of proteins. The N-linked glycans are usually seen in highly branched complex structures and covalently added on asparagine residues during N-linked glycosylation by endoplasmic reticulum-resident GTs. The O-linked glycans are often added on serine or threonine residues during O-linked glycosylation by Golgi complex-resident GTs.

Gal: Galactose; GalNAc: N-Acetylgalactosamine; Glc: Glucose; GlcNAc: N-Acetylglucosamine; GT: Glycosyltransferase; Man: Mannose.

- N-linked glycans are commonly attached to **asparagine**, whereas O-linked glycans are commonly attached to **serine or threonine**.
- Detailed glycan chemistry is beyond the scope of this introductory lecture.

7-6. Glial Cells Play Diverse Roles in Neural Function

Neuronal function depends on glial specialization

Major glial cell types

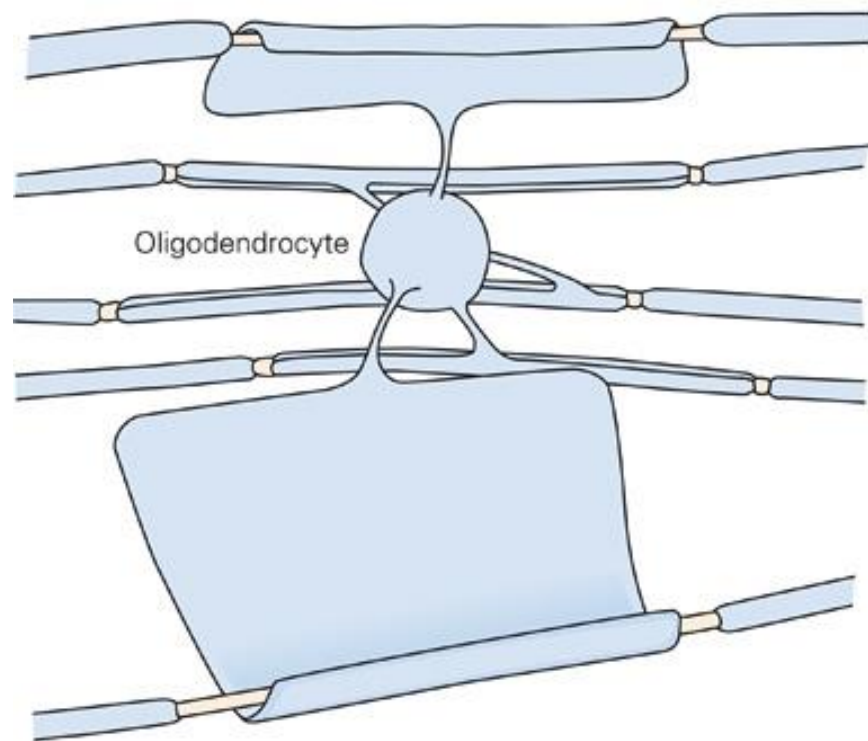
Glia are not just “support cells.” They actively shape conduction, synaptic signaling, homeostasis, and immune responses.

Oligodendrocytes	CNS myelin; one cell can myelinate multiple axons
Schwann cells	PNS myelin; one cell myelinates one axonal segment
Astrocytes	Ion/neurotransmitter homeostasis, synaptic modulation, blood-brain barrier (BBB) support
Microglia	Resident immune cells; surveillance, synaptic pruning, injury response
Ependyma/choroid plexus	Cerebrospinal fluid (CSF) movement and production

Myelinating glia: CNS versus PNS

- Oligodendrocytes and Schwann cells produce myelin.
- CNS: **oligodendrocytes** can myelinate segments of many axons.
- PNS: one **Schwann cell** forms myelin around one segment of one axon.
- Myelin is a lipid-rich membrane that electrically insulates axons.

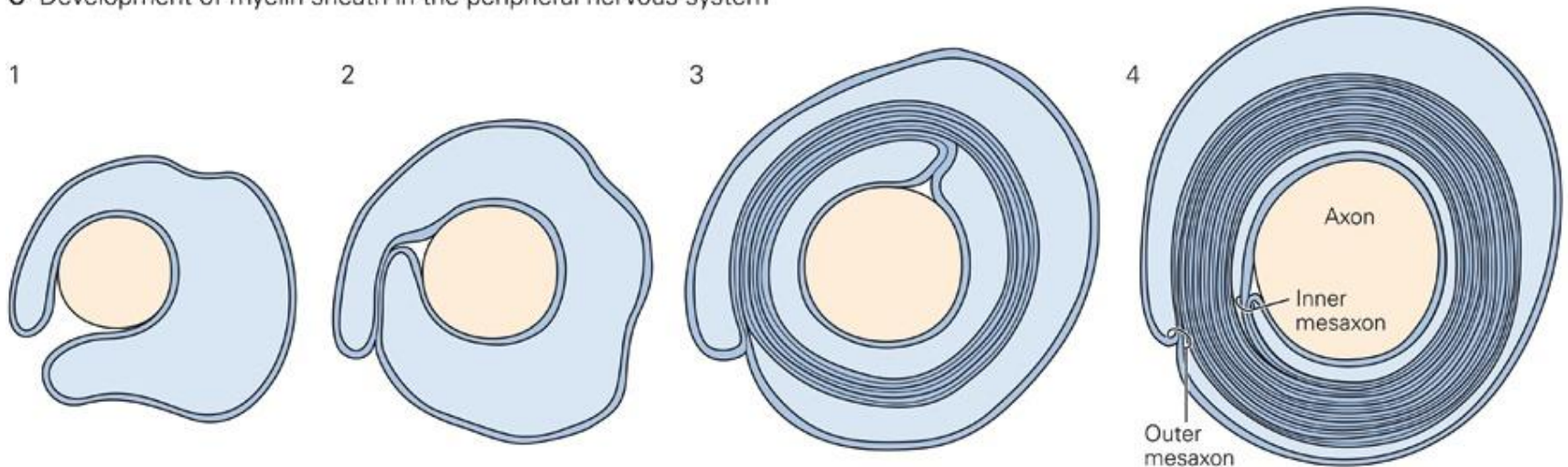
A Myelination in the central nervous system



How myelin forms: the concept, not the terminology

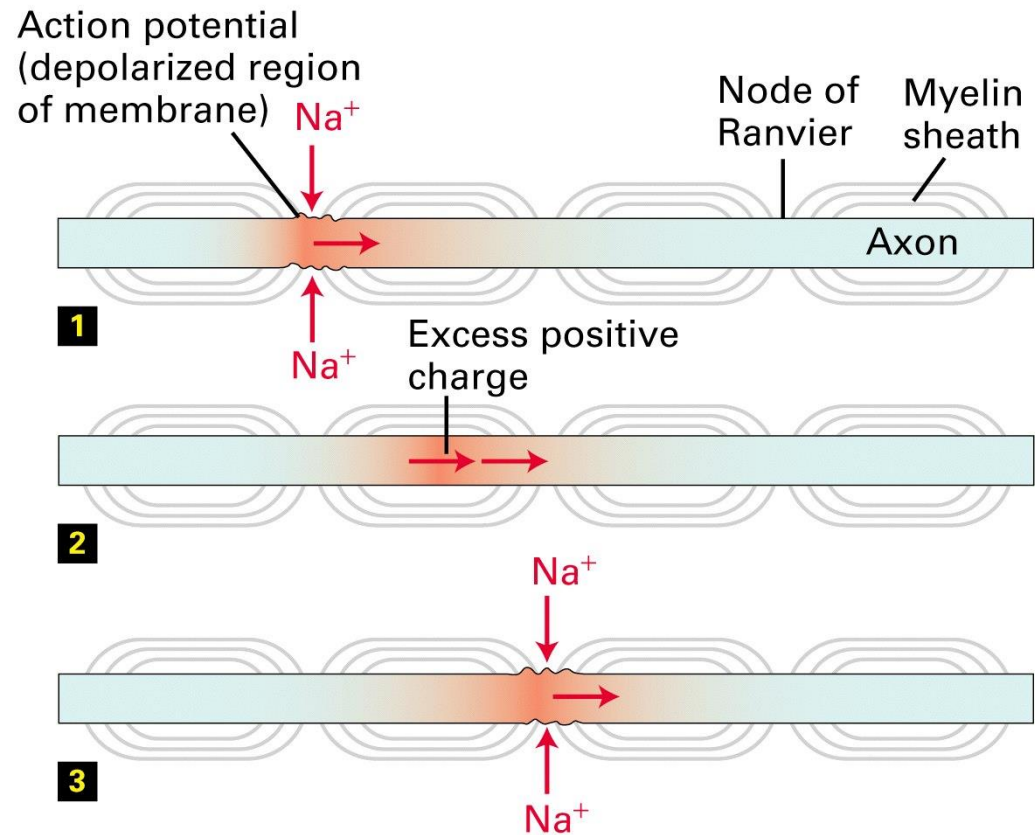
- A glial membrane wraps repeatedly around an axon.
- Layers become compacted to form an insulating sheath.

C Development of myelin sheath in the peripheral nervous system



Myelin increases conduction velocity

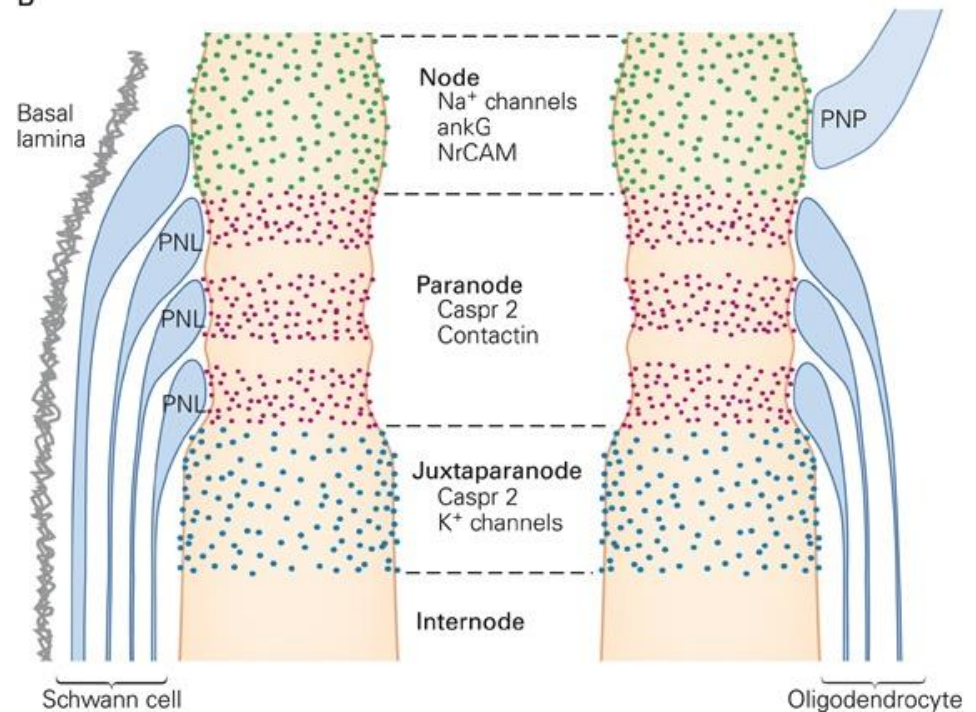
1. Myelin reduces current leakage across the axonal membrane.
2. **Nodes of Ranvier** are gaps where the axonal membrane is exposed.
3. Voltage-gated Na^+ channels are concentrated at nodes.
4. The action potential effectively “jumps” from node to node.



saltatory conduction

Nodes of Ranvier: keep the principle, not every molecule

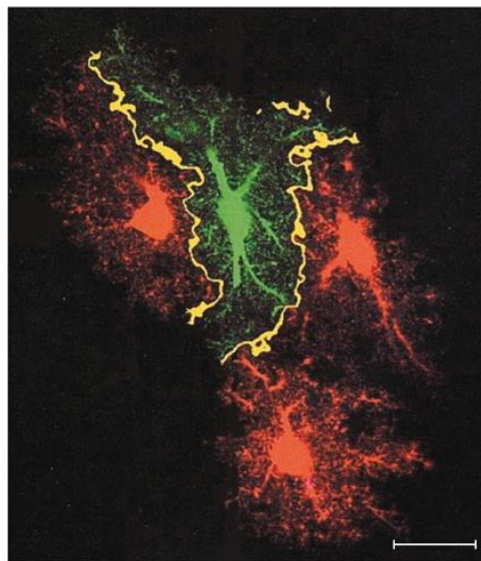
- Nodes are regularly spaced interruptions in the myelin sheath.
- They organize ion channels for efficient impulse propagation.
- Molecular details such as Caspr2, ankyrin G, NrCAM, and paranodal loops are advanced details, just for your own reference.



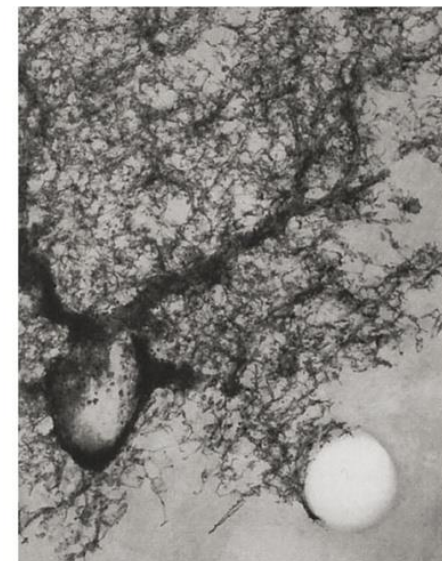
Astrocytes support synaptic signaling

- Astrocytes are abundant CNS glia with highly branched processes.
- Their processes contact synapses and blood vessels.
- They regulate extracellular ions, neurotransmitters, and metabolic support.
- Their functions make them active partners of neurons.

A



B



Major astrocyte functions

- K^+ buffering: stabilizes extracellular ionic environment.
- Glutamate uptake: limits excitotoxicity and supports neurotransmitter cycling.
- Synaptic modulation: astrocytes respond to neuronal activity.
- Synapse formation, trophic support, oxidative protection.
- BBB support: astrocyte end-feet interact with cerebral vessels.

Astrocytes buffer extracellular K^+

Problem

- Active neurons release K^+ into the extracellular space.
- Excess extracellular K^+ changes neuronal excitability.

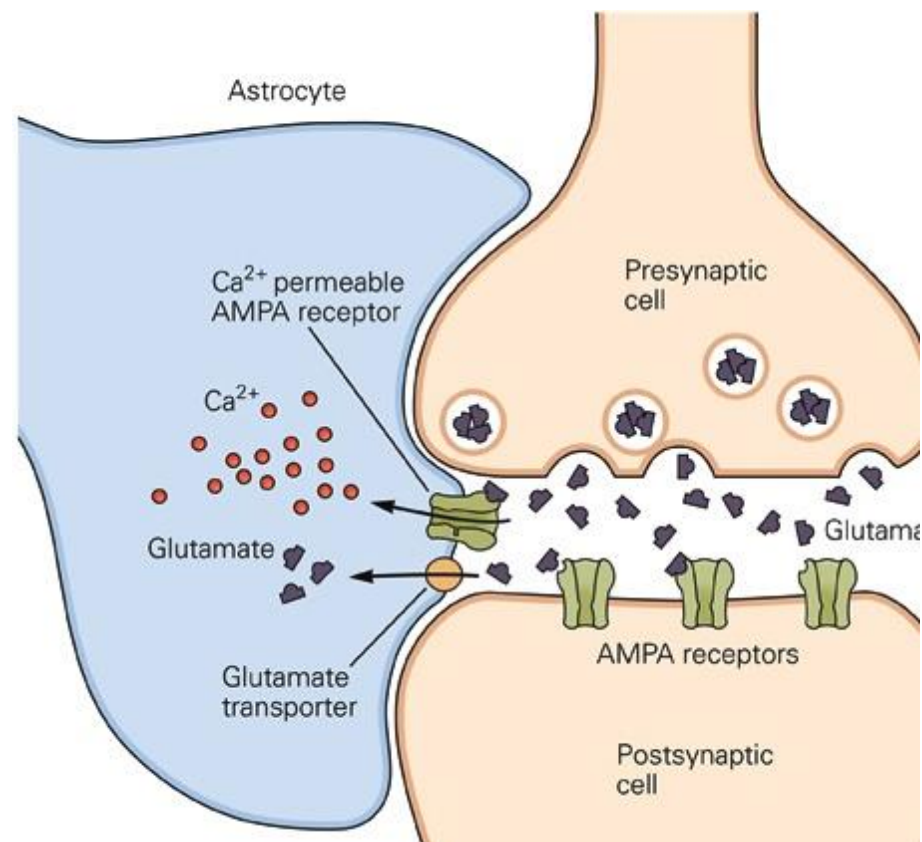
Astrocyte response

- Astrocytes take up K^+ near active synapses.
- They redistribute ions through astrocytic networks and contacts with blood vessels.

Physiological meaning: astrocytes help keep the extracellular environment stable while neurons fire.

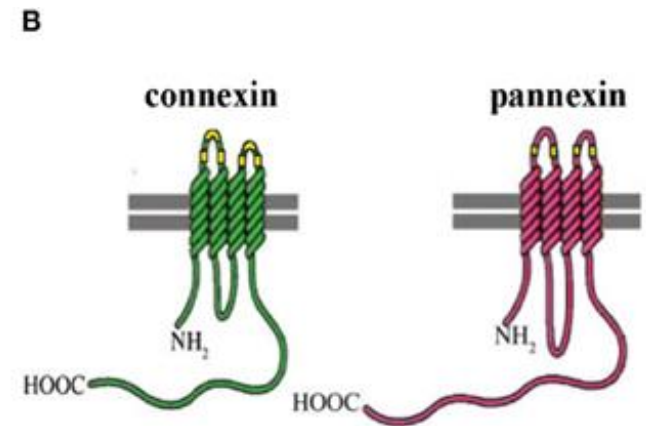
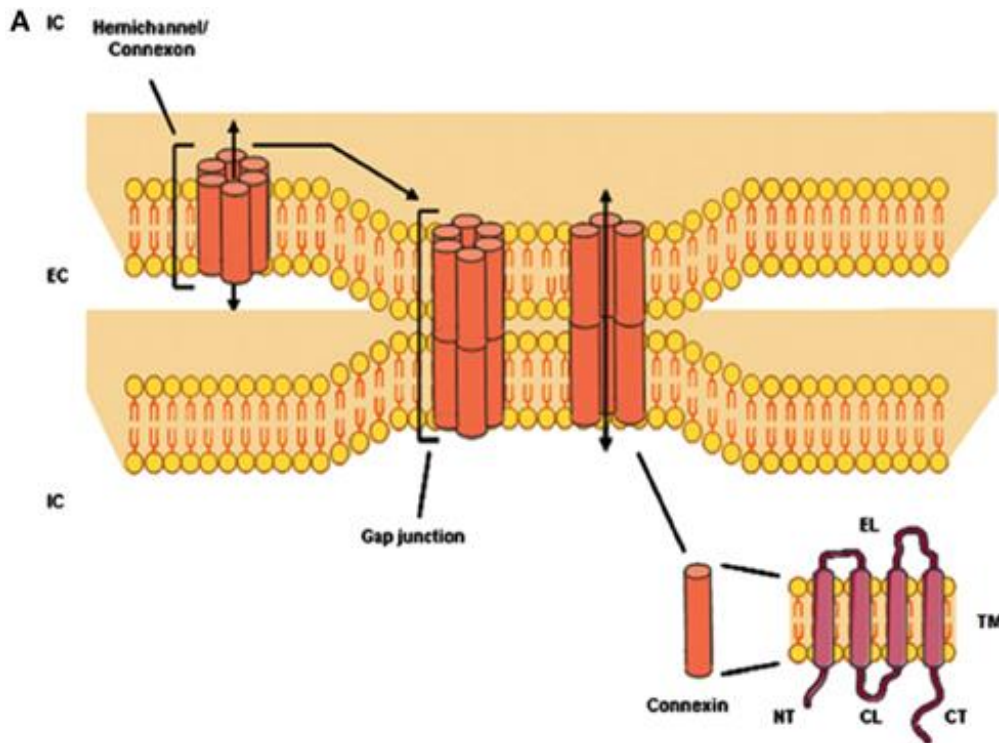
Astrocytes clear glutamate from the synaptic cleft

- Astrocytes express high-affinity **glutamate transporters**.
- Uptake of glutamate in synaptic cleft lowers extracellular glutamate and limits excitotoxicity.
- Inside astrocytes, glutamate is converted to **glutamine**, which can be returned to neurons.



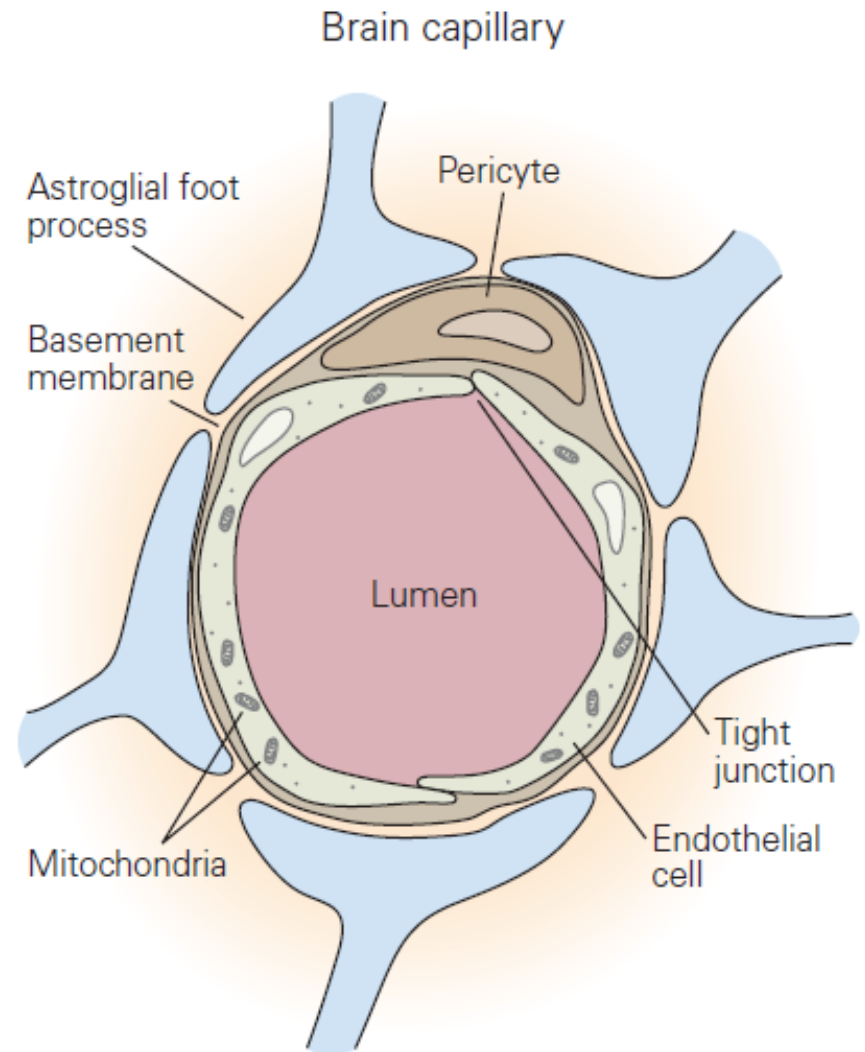
Astrocytic Ca^{2+} signaling and gap junctions

- Astrocytes can respond to neuronal activity with intracellular Ca^{2+} signals.
- Adjacent astrocytes are coupled through **gap junctions**.
- This provides a way for astrocyte networks to coordinate local signaling.



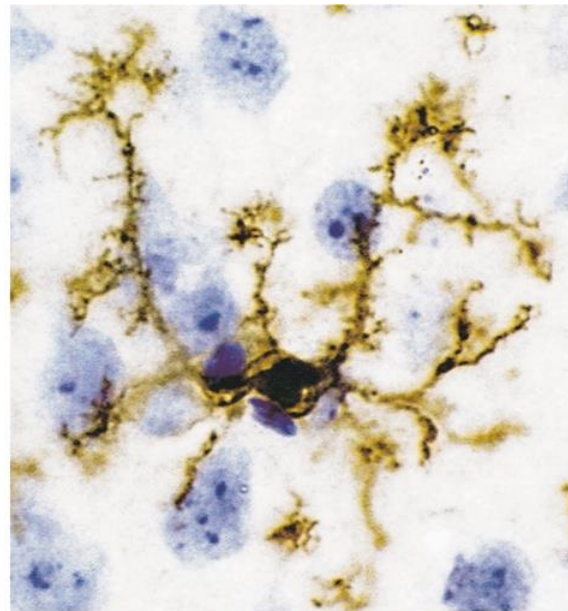
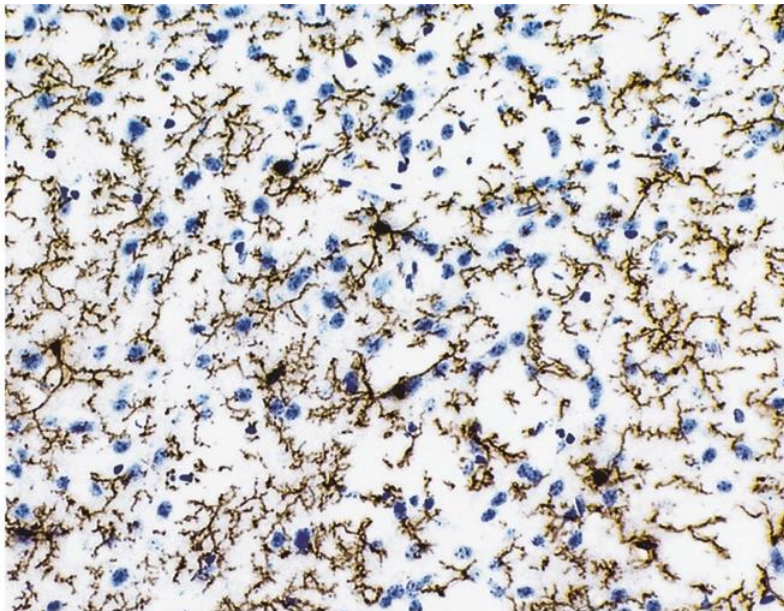
Astrocytes and the blood–brain barrier (BBB)

- The major physical barrier is formed by tight junctions between **cerebral endothelial cells**.
- Astrocytic end-feet surround capillaries and help maintain BBB properties.
- Pericytes and basement membrane are also part of the neurovascular unit.
- BBB transport is selective—not simply an open diffusion path.



Microglia: resident immune cells of the CNS

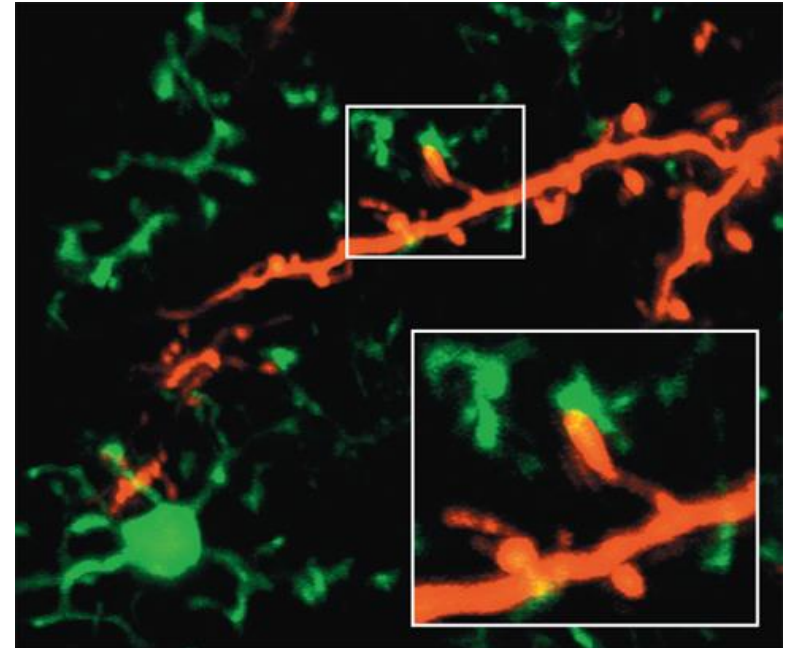
- Microglia are CNS-resident immune cells and phagocytes.
- Microglia do not arise from the **neuroectodermal lineage**. Rather, they originate from **yolk sac myeloid progenitors** and colonize the brain early in development.
- Microglia remain dynamic even in the healthy brain. They continuously survey neural tissue.



Branched processes allow microglia to monitor their local environment and contact neurons, synapses, and other glial cells.

Microglia in development, injury, and disease

- Development: help sculpt circuits by engulfing synaptic elements.
- Injury: rapidly change morphology, gene expression, and process motility.
- Protective roles: phagocytose debris, dying cells, and toxic protein aggregates.
- Pathological roles: excessive inflammatory signaling can contribute to synapse loss and neurodegeneration.



Two-photon imaging in the olfactory bulb of adult mice shows microglial processes (green) connecting to neurons (red).

Microglial activation states are diverse: context-dependent responses rather than fixed binary identities

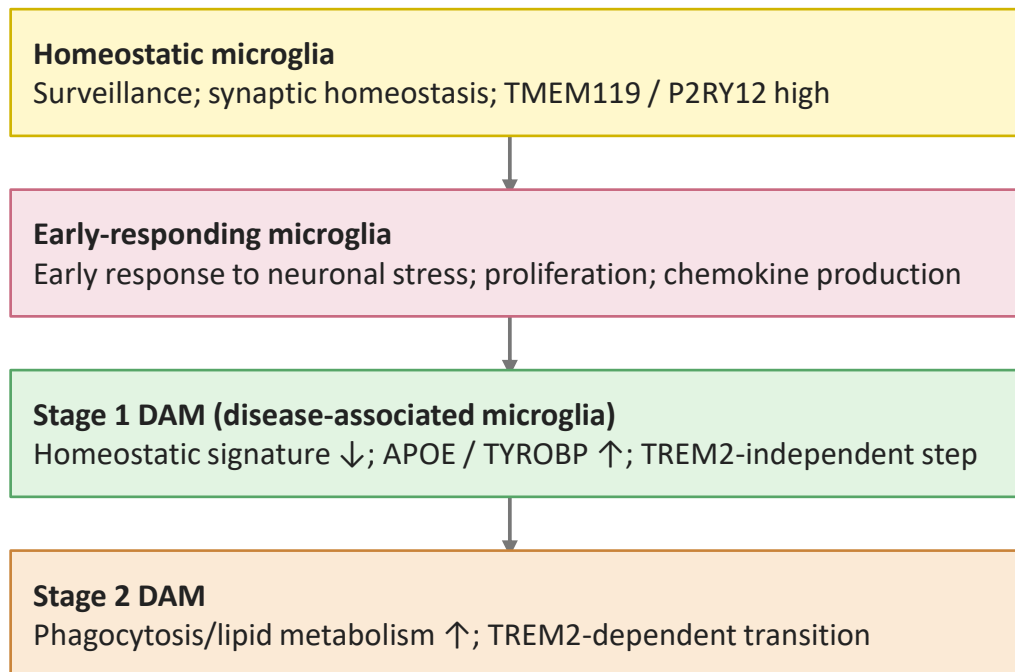
Historical M1/M2 model

M1: pro-inflammatory/neurotoxic

M2: anti-inflammatory/neuroprotective

Such a binary classification is now oversimplified.

Example from AD/neurodegeneration models: disease-associated states



Microglia shift among multiple context-dependent states; DAM is an illustrative framework from AD/neurodegeneration research, not a universal linear pathway.

Ependymal cells, choroid plexus, and CSF

- Ependymal cells line the ventricles; cilia help move CSF.
- Choroid plexus epithelium produces most CSF.
- CSF circulates through the ventricular system and surrounding CNS.
- CSF composition is regulated by epithelial transporters and channels.

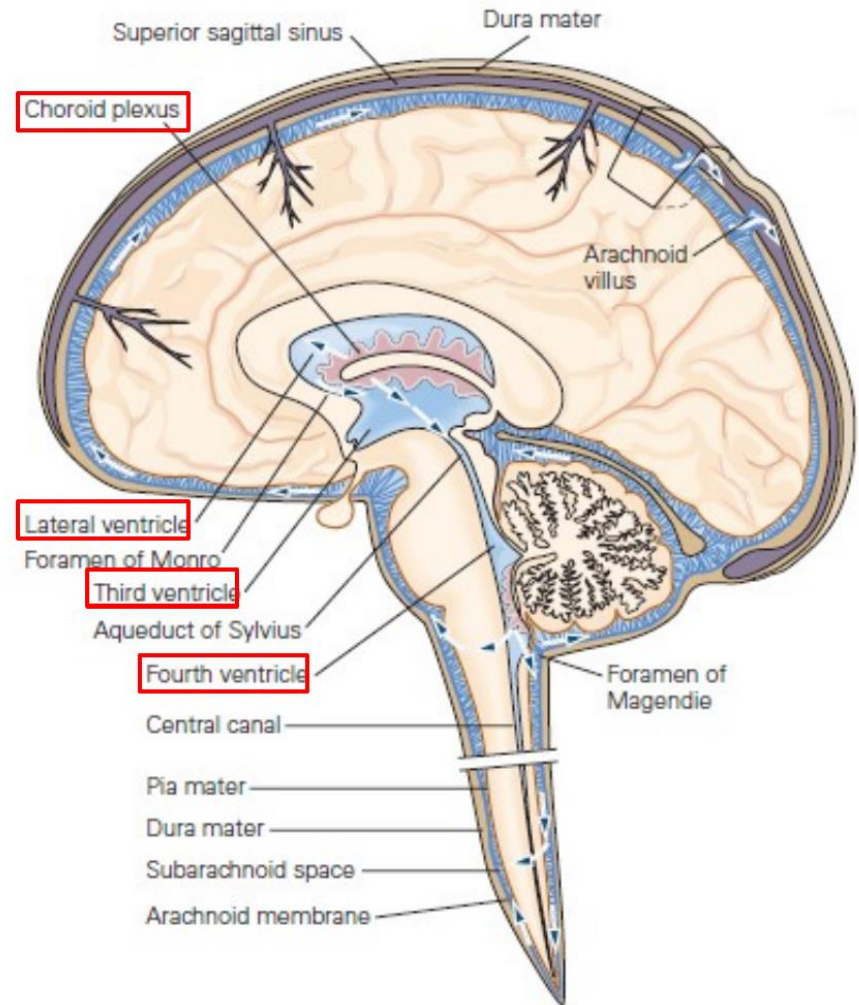


Figure D-8 Distribution of cerebrospinal fluid.

Question or Discussion?

Ding-I Yang, Ph.D.

Campus phone #: 67386

E-mail: diyang@nycu.edu.tw